

Heart Failure: Epidemiology, Pathophysiology, Patient Evaluation, and Guideline-Directed Management

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Session Objectives

1. Describe the risk factors for developing heart failure.
2. Identify types and stages of heart failure: A → D; HFrEF, HFpEF; right- and left-sided heart failure, acute and chronic heart failure.
3. Apply and interpret appropriate tests for heart failure diagnosis.
4. Review the clinical manifestations of heart failure.
5. Apply guideline-directed medical therapy (pharmacologic and non-pharmacologic) in management of heart failure across the spectrum of stages and classes.
6. Explain the importance of improving care coordination amongst interprofessional team members to decrease mortality of patients with heart failure.



8. Patient Presentations Related to the Circulatory and Hematologic Systems

8.6 Shortness of Breath/Dyspnea and Orthopnea

- blood pressure abnormalities – diastolic or systolic cardiac dysfunction

PHYSICAL EXAM FINDINGS • cardiac auscultation, murmurs, and heart sounds • cardiac percussion and palpation (eg, thrills, point of maximal impulse, parasternal heave) • hemodynamic assessment (eg, blood pressure measurement, tilt test, jugular venous pressure, pulses, edema)

LABORATORY TEST FINDINGS AND DIAGNOSTIC IMAGING ECG, 12-lead and rhythm strips

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Cardiovascular System

Organ structure and function chambers, valves cardiac cycle, mechanics, heart sounds, cardiac conduction hemodynamics, including blood volume and systemic vascular resistance circulation in specific vascular beds, including pulmonary and coronary

Abnormal Processes: Health and Health Maintenance, Screening, Diagnosis, Management, Risks, Prognosis

Heart failure: chordae tendineae rupture; congestive heart failure; cor pulmonale; diastolic dysfunction; systolic dysfunction; mitral valve dysfunction; heart failure secondary to myocardial infarction; high-output heart failure, including thyrotoxicosis-induced, anemia-induced; tachycardia-induced; cardiogenic pulmonary edema

Case

- A 65-year-old man with a history of HTN, hyperlipidemia, and tobacco use reports to the clinic because of 2 weeks of increasing dyspnea on exertion. He now sleeps on 4 pillows at night to aid in breathing. He has T2DM variably controlled with metformin and diet, and drinks 3-4 glasses of wine/day. Physical examination: jugular venous distention, an S₃ heart sound, and bilateral LE edema. ECG shows Q waves in leads V₂-V₄. He gives a history of hospitalization for "issues with my heart" about a year ago when he had sudden chest pain that radiated to his left arm, after which he had been asymptomatic.
- Which of the following is the most likely cause of this patient's condition?
 - A. Alcohol
 - B. Aortic stenosis
 - C. Coronary artery disease
 - D. Coronavirus
 - E. Coxsackievirus B
 - F. Sarcomere gene mutation



Heart Failure is a Clinical Syndrome

- Heart is unable to pump blood at a rate sufficient to meet metabolic demands leading to symptoms of fatigue, shortness of breath and volume overload
- Abnormalities in cardiac function and neurohormonal regulation, which are accompanied by effort intolerance, fluid retention, and a reduced longevity
- Complex clinical syndrome that can result from any structural or functional cardiac disorder that impairs the ability of ventricle to fill or pump

Epidemiology

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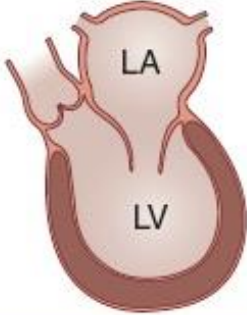
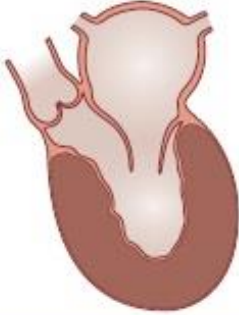


- 6.7 million adults in US
- Higher incidence and prevalence among Black individuals
- More younger patients
- 60 million worldwide
- 50% mortality within 5 years of diagnosis
- #1 cause for hospital admission in pts over 65
- Higher mortality in rural areas

https://www.cdc.gov/heartdisease/heart_failure.htm

Bozkurt B, Ahmad T, Alexander K, Baker WL, Bosak K, Breathett K, Carter S, Drazner MH, Dunlay SM, Fonarow GC, Greene SJ, Heidenreich P, Ho JE, Hsich E, Ibrahim NE, Jones LM, Khan SS, Khazanie P, Koelling T, Lee CS, Morris AA, Page RL 2nd, Pandey A, Piano MR, Sandhu AT, Stehlik J, Stevenson LW, Teerlink J, Vest AR, Yancy C, Ziaeian B; WRITING COMMITTEE MEMBERS. HF STATS 2024: Heart Failure Epidemiology and Outcomes Statistics An Updated 2024 Report from the Heart Failure Society of America. J Card Fail. 2025 Jan;31(1):66-116. doi: 10.1016/j.cardfail.2024.07.001.

Types of Heart Failure: by Structure

	Dilated Cardiomyopathy	Hypertrophic Cardiomyopathy	Restrictive Cardiomyopathy
Ventricular morphology	Dilated LV with little concentric hypertrophy 	Marked hypertrophy, often asymmetric 	Fibrotic or infiltrated myocardium 
Etiology examples	Genetic, inflammatory, toxic, metabolic, peripartum	Genetic	Amyloidosis, genetic
Symptoms	Fatigue, weakness, dyspnea, orthopnea, PND	Dyspnea, angina, syncope	Dyspnea, fatigue
Physical exam	Pulmonary crackles, S3; if RV failure present: JVD, hepatomegaly, peripheral edema	S4; if outflow obstruction present: systolic murmur loudest at the left sternal border, accompanied by mitral regurgitation	Predominantly signs of RV failure: JVD, hepatomegaly, peripheral edema
Pathophysiology	Impaired systolic contraction	Impaired diastolic relaxation; LV systolic function vigorous, often with dynamic obstruction	"Stiff" LV with impaired diastolic relaxation but usually normal systolic function
Cardiac size on chest radiograph	Enlarged	Normal or enlarged	Usually normal
Echocardiogram	Dilated, poorly contractile LV	LV hypertrophy, often more pronounced at the septum; systolic anterior movement of MV with mitral regurgitation	Usually normal systolic contraction; the myocardium may have enhanced echogenicity in infiltrative disorders

LV, left ventricle; PND, paroxysmal nocturnal dyspnea; RV, right ventricle; JVD, jugular venous distention; MV, mitral valve.

Types of Heart Failure: Left vs. Right

Left-side Heart Failure:

- pump failure:
 - HFrEF and HFpEF

Right-side heart failure:

- Most commonly caused by left-sided HF
 - Cor pulmonale: isolated right heart failure, infrequent, secondary to lung pathology

Classification of HF by Left Ventricular function

- Left-sided heart failure can be divided into HFrEF (systolic) and HFpEF (diastolic) failure
- Normal EF= 55-70%

HF with reduced EF (HFrEF):

- HF with LVEF $\leq 40\%$

HF with mildly reduced EF (HFmrEF):

- HF with LVEF 41-49%

HF with preserved EF (HFpEF):

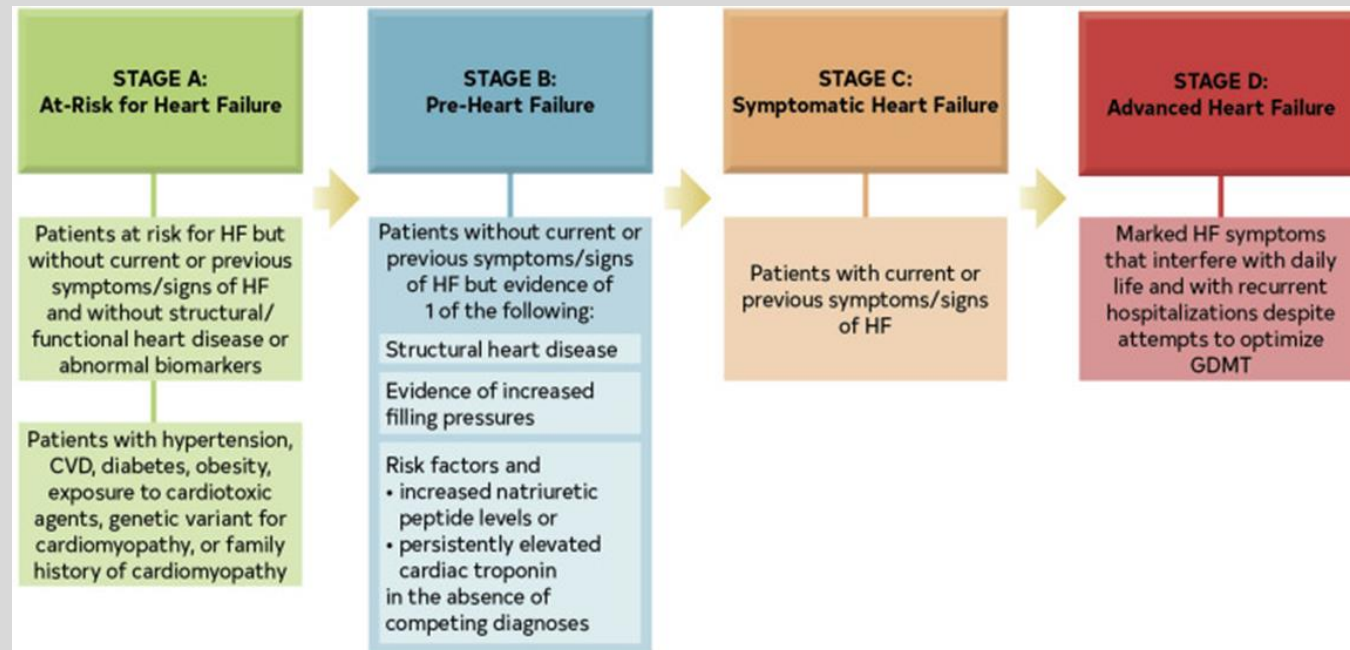
- HF with LVEF $\geq 50\%$

HF with improved EF (HFimpEF):

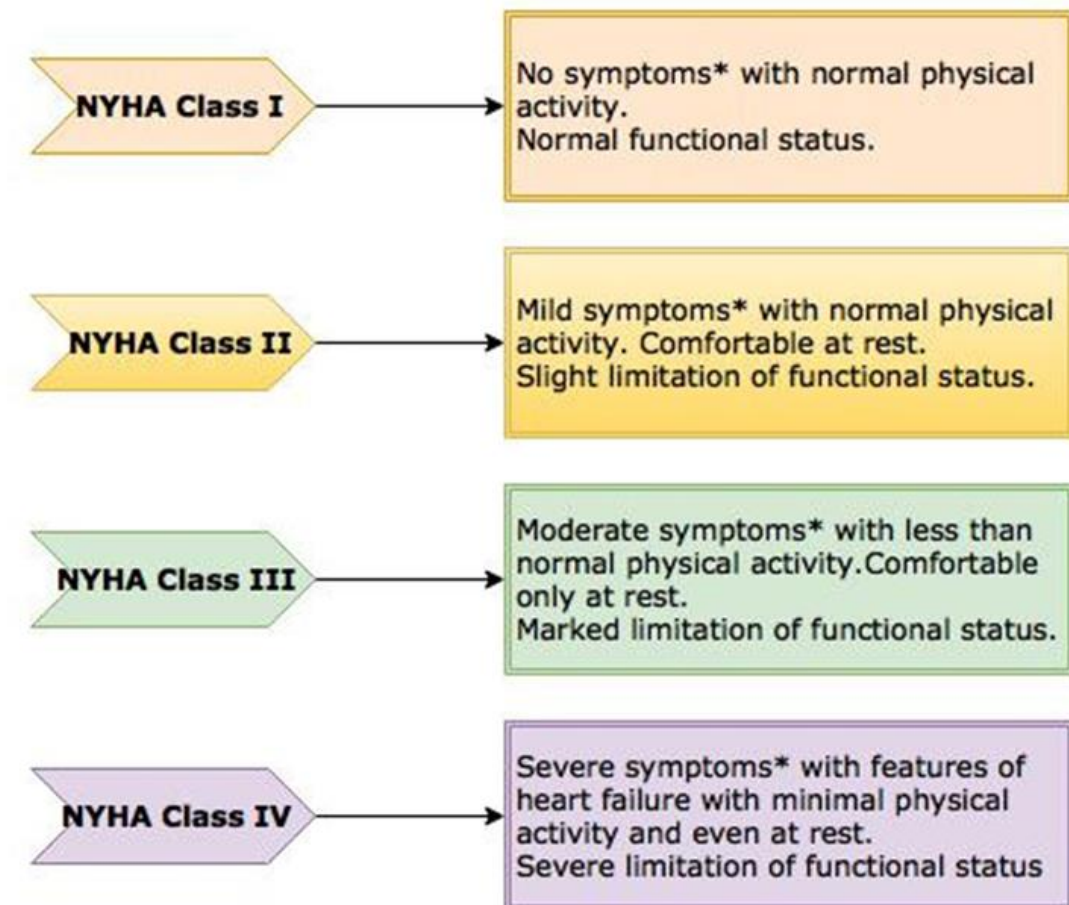
- HF with a baseline LVEF $\leq 40\%$, a ≥ 10 point increase from baseline LVEF, and a second measurement of LVEF $> 40\%$

Types of HF: Functional Classification

ACC/AHA Stages of HF



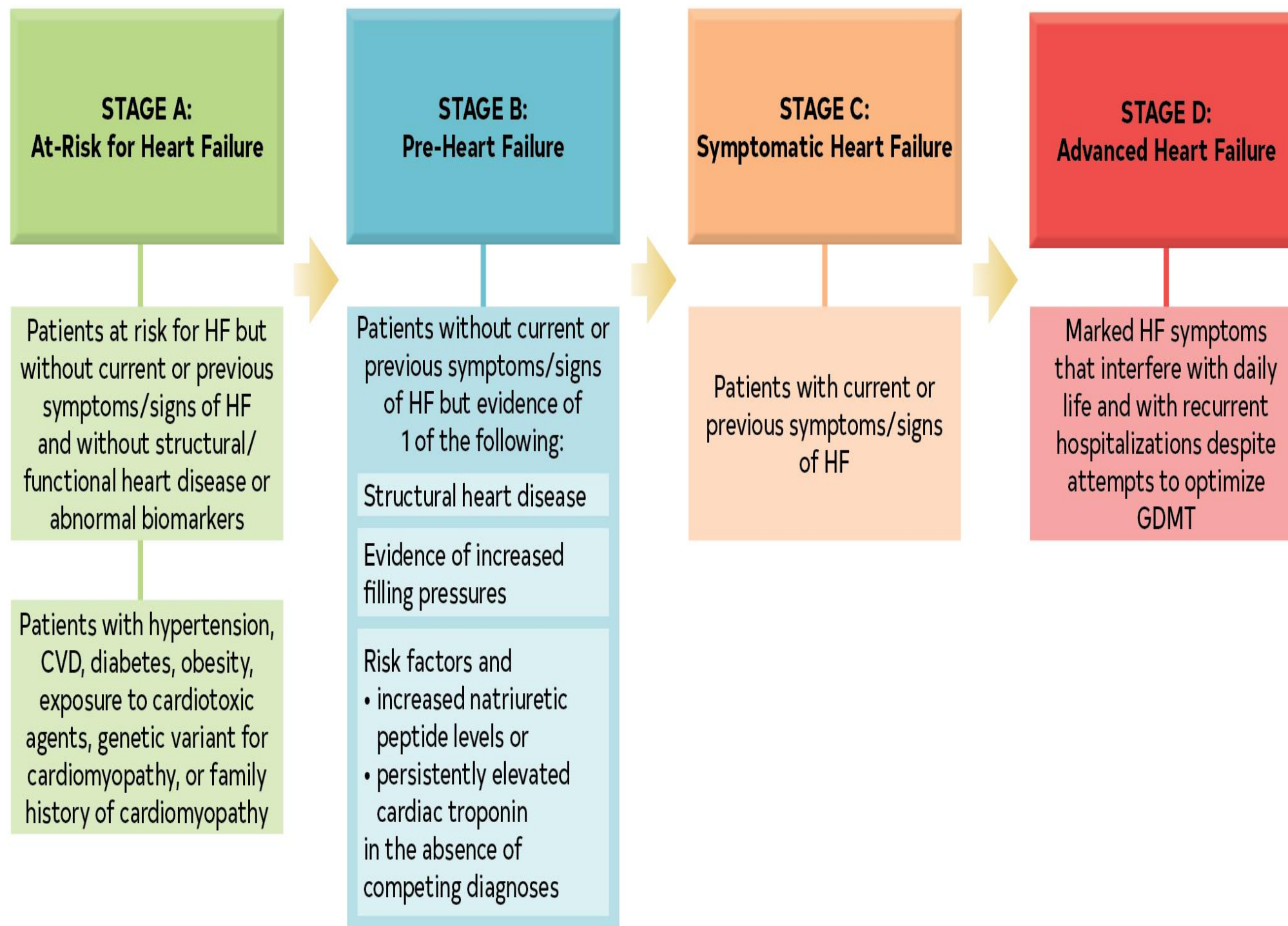
New York Heart Association (NYHA) Classification of severity of Heart Failure



Symptoms - Fatigue, palpitations, chest pain, dyspnea, syncope

ACC/AHA Stages of HF

ACC indicates American College of Cardiology; AHA, American Heart Association; CVD, cardiovascular disease; GDMT, guideline-directed medical therapy; and HF, heart failure.



Causes of HF

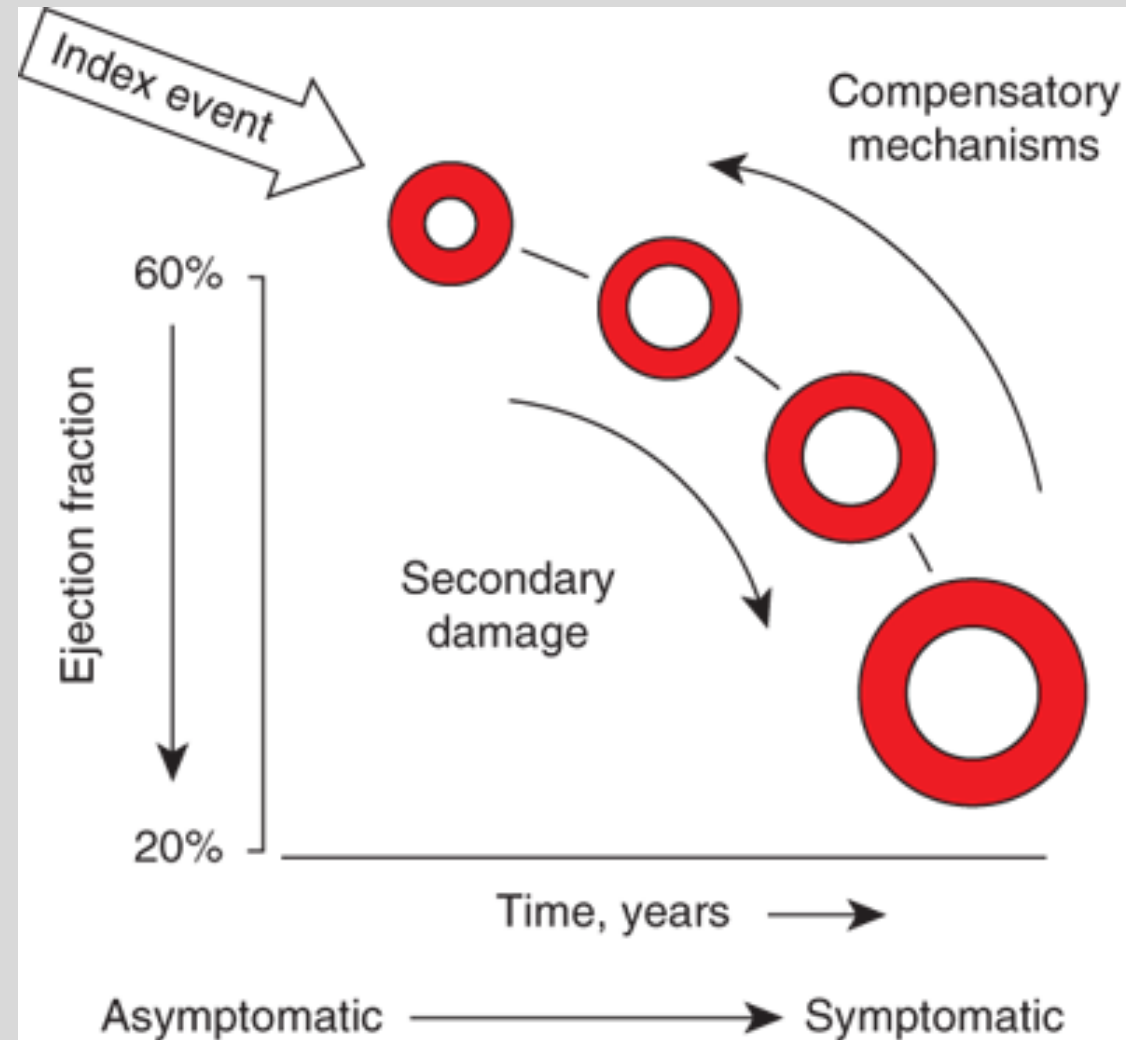


Harrison's Principles of Internal Medicine, 21e >Heart Failure: Pathophysiology and Diagnosis

Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson+
TABLE 257-2Selected Causes of Heart Failure

Heart Failure with Reduced Ejection Fraction	
Coronary artery disease	Nonischemic cardiomyopathy
Myocardial infarction	Infiltrative disorders
Myocardial ischemia	Familial disorders
	Tachycardia induced
Valvular heart disease	Toxic cardiomyopathy
Aortic stenosis or regurgitation	Chemotherapy, immunotherapy
Mitral or tricuspid regurgitation	Drugs such as hydroxychloroquine
	Alcohol, cocaine
Congenital heart disease	Chronic lung/pulmonary vascular disease
Intracardiac shunts	Cor pulmonale
Repaired defects	Pulmonary arterial hypertension
Systemic right ventricular failure	
Infectious	Autoimmune disease
Chagas	Giant cell myocarditis
HIV	Lupus myocarditis
Heart Failure with Preserved Ejection Fraction	
Hypertension	Coronary artery disease
Valvular heart disease	Restrictive cardiomyopathy
Aortic stenosis	Amyloidosis
Mitral stenosis	Sarcoidosis
	Hemochromatosis
	Glycogen storage disease
Hypertrophic cardiomyopathy	Radiation therapy
Constrictive pericarditis	Aging

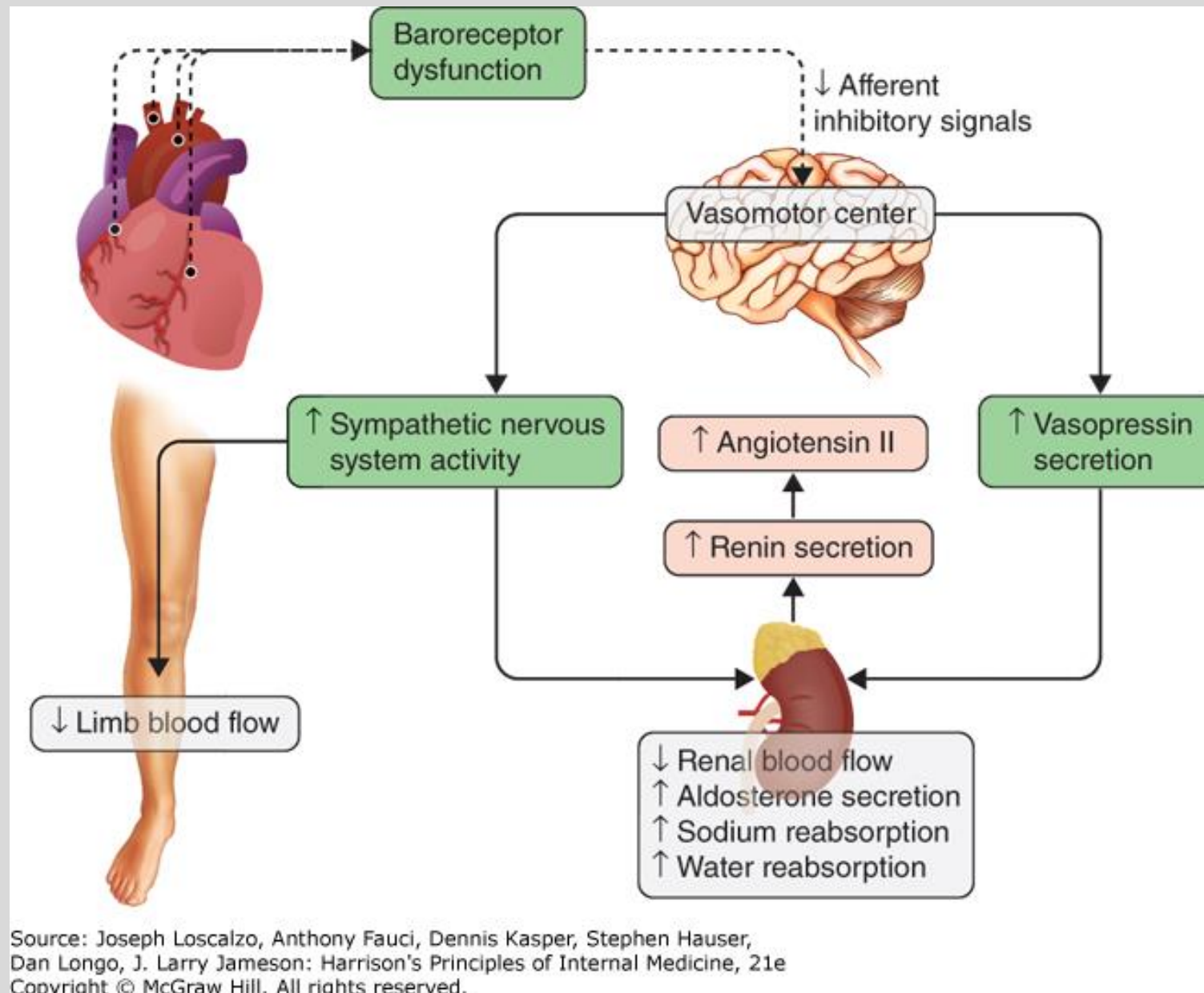
Pathophysiology of HFrEF



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition Copyright © McGraw-Hill Education. All rights reserved.

- Inciting event (MI; pressure or volume overload; genetic cardiomyopathy)
- Activation of compensatory mechanisms
- Decline in the heart's pumping capacity

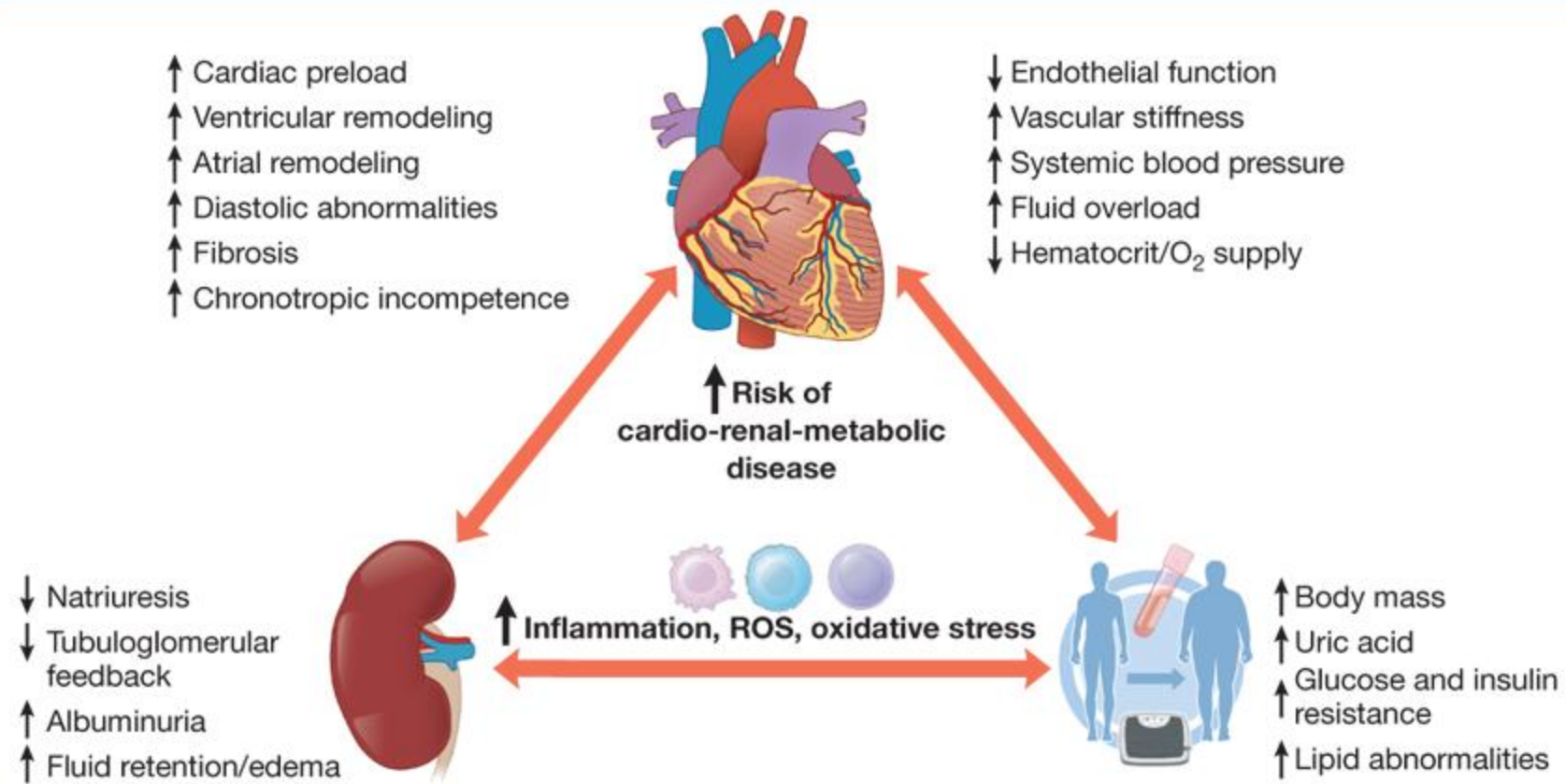
Pathophysiology: Neurohormonal Activation



- $BP = CO \times SVR$
 - Adrenergic Nervous System
 - Renin-Angiotensin-Aldosterone System
 - Increased production of Antidiuretic Hormone (ADH)

Pathophysiology of HFpEF

Heart failure with preserved ejection fraction: multiorgan systemic pathophysiology



- Impaired diastolic relaxation of the LV
- Vascular stiffness
- Renal dysfunction
- Metabolic inflammation
- Oxidative stress and impaired nitric oxide signaling

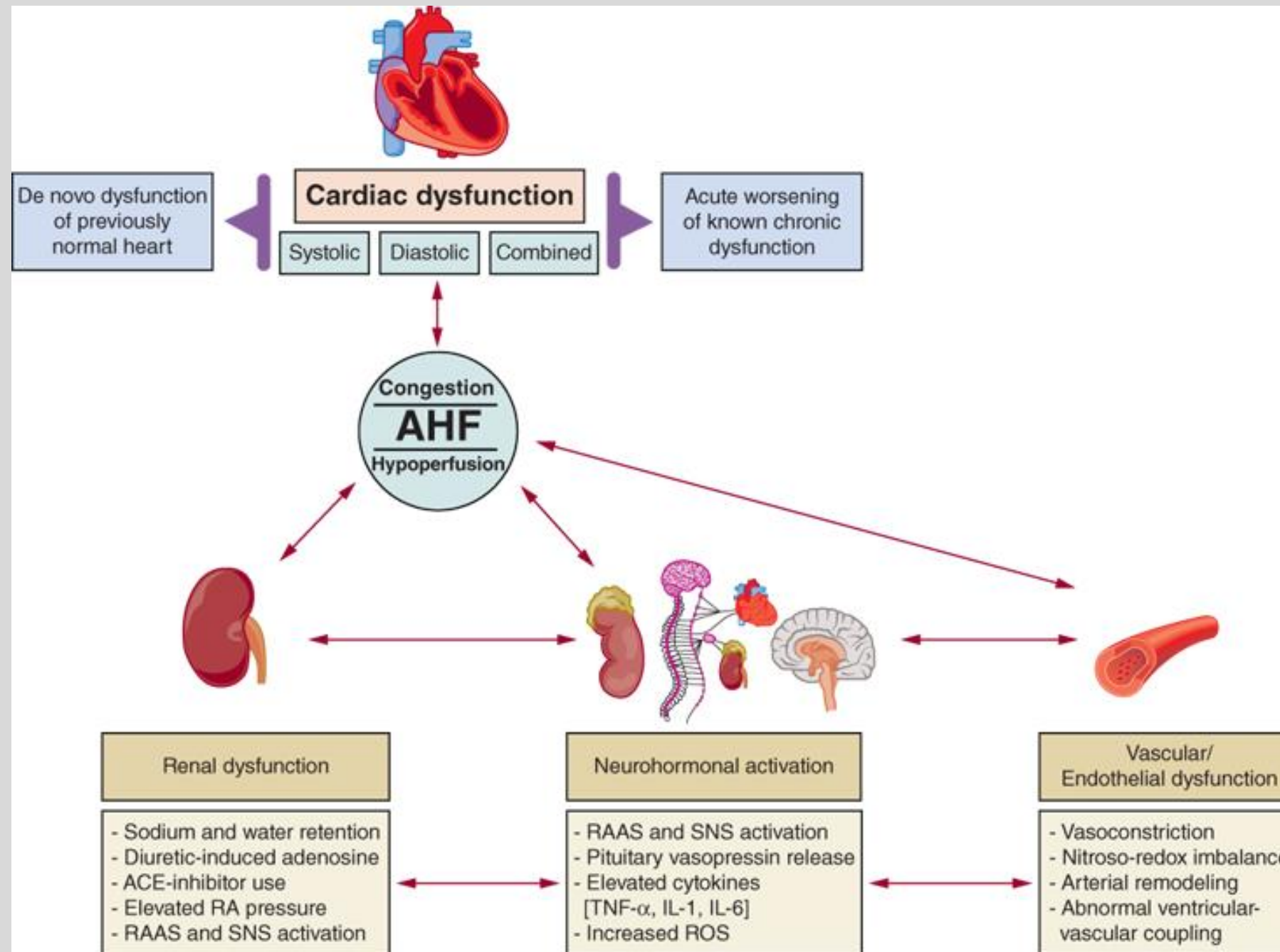
Chapter 49 Fuster and Hurst's Central Illustration. Patients with heart failure with preserved ejection fraction have increased risk of cardio-renal-metabolic disease. ROS, reactive oxygen species.

Source: V. Fuster, J. Narula, P. Vaishnava, M.B. Leon, D.J. Callans, J. Rumsfeld, A. Poppas: Fuster and Hurst's The Heart, 15e
Copyright © McGraw Hill Education. All rights reserved.

Ferreira J, Sharma K, Butler J. Diagnosis and Management of Heart Failure with Preserved Ejection Fraction. In: Fuster V, Narula J, Vaishnava P, Leon MB, Callans DJ, Rumsfeld J, Poppas A. eds. *Fuster and Hurst's The Heart*, 15e. McGraw Hill; 2022.

Givertz MM, Mehra MR. Heart Failure: Pathophysiology and Diagnosis. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson J. eds. *Harrison's Principles of Internal Medicine*, 21e. McGraw-Hill Education; 2022.

Pathophysiology: Acute HF



Source: V. Fuster, J. Narula, P. Vaishnava, M.B. Leon, D.J. Callans, J. Rumsfeld, A. Poppas: Fuster and Hurst's The Heart, 15e
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Clinical Manifestations

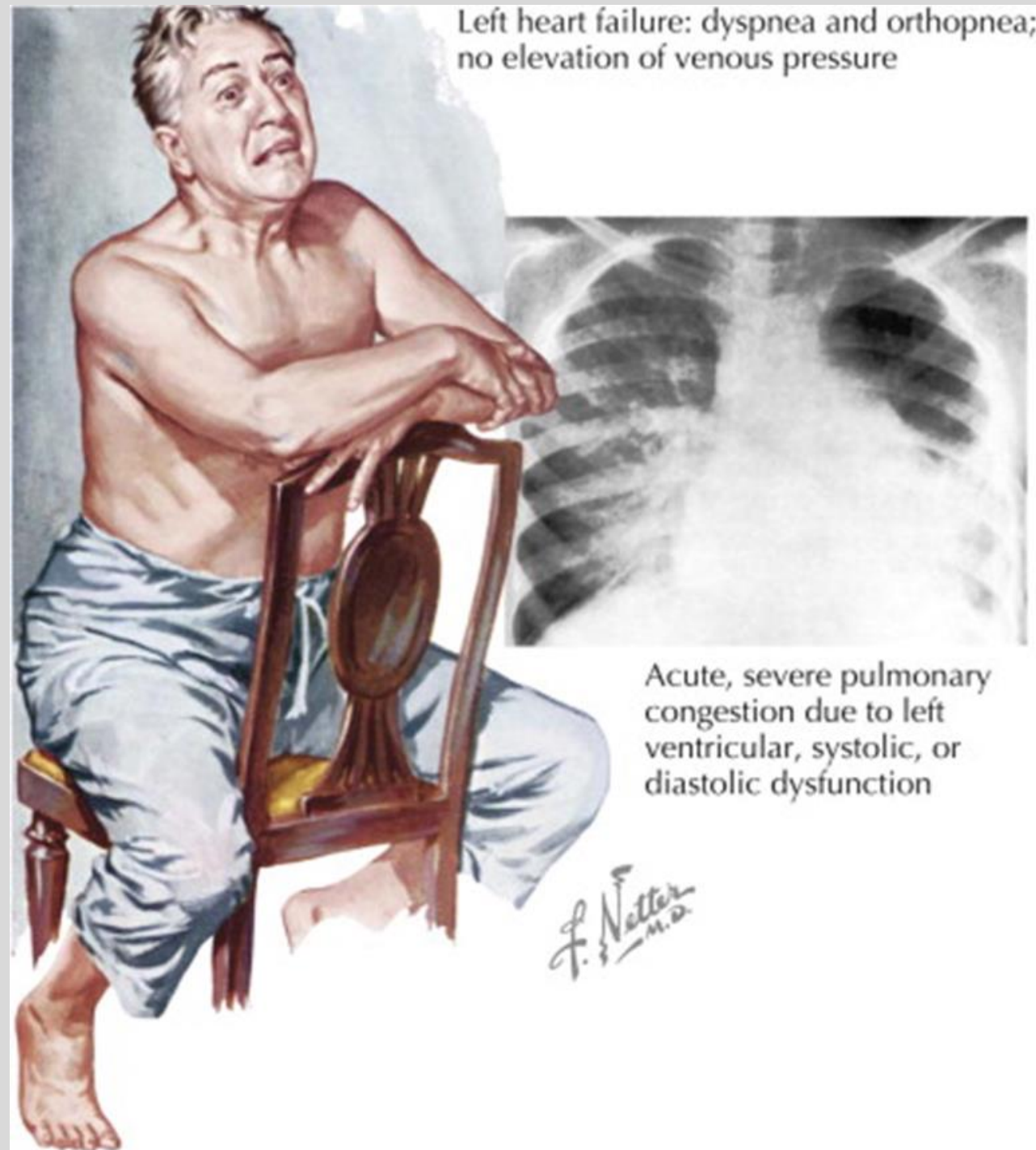
- Dyspnea
- Fatigue
- Exercise intolerance
- Cough
- Orthopnea
- Paroxysmal Nocturnal Dyspnea (PND)
- Cheyne-Stokes Respiration
- Acute pulmonary edema
- Abnormal cardiovascular exam (HTN, murmur, elevated JVP, pulmonary crackles, S3)
- Lower extremity edema



Left-sided HF: Pathophysiology & Clinical Manifestations

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- Pump failure, “forward failure”:
- Elevated pressure and congestion in the pulmonary veins and capillaries:
 - Orthopnea
 - Dyspnea
 - PND

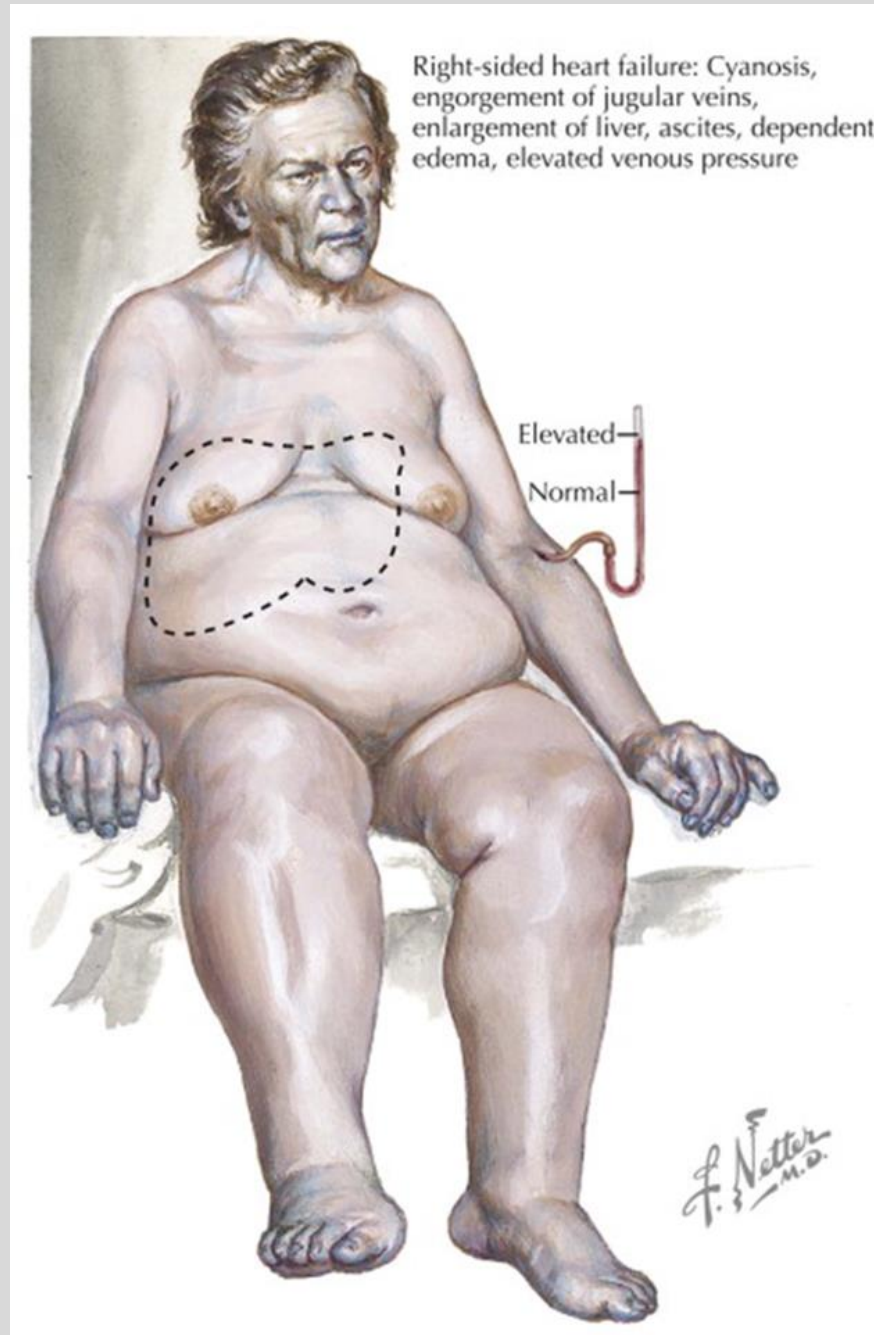
Chang P. (2010) Netter's Cardiology, 18, 145-153 Retrieved from <https://www.clinicalkey.com>

Francis GS, Tang W, Walsh RA. Francis G.S., Tang W, Walsh R.A. Francis, Gary S., et al. Chapter 26. Pathophysiology of Heart Failure. In: Fuster V, Walsh RA, Harrington RA. Fuster V, Walsh R.A., Harrington R.A. Eds. Valentin Fuster, et al.eds. Hurst's The Heart, 13e. New York, NY: McGraw-Hill; 2011.

Right-sided HF: Pathophysiology & Clinical Manifestations

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“Backward heart failure”:

- abnormal function of the LV overloads the RV from increased pulmonary pressures:
 - Jugular venous distention
 - Hepatomegaly
 - Peripheral edema

Chang P. (2010) Netter's Cardiology, 18, 145-153 Retrieved from <https://www.clinicalkey.com>

Francis GS, Tang W, Walsh RA. Francis G.S., Tang W, Walsh R.A. Francis, Gary S., et al. Chapter 26. Pathophysiology of Heart Failure. In: Fuster V, Walsh RA, Harrington RA. Fuster V, Walsh R.A., Harrington R.A. Eds. Valentin Fuster, et al. eds. Hurst's The Heart, 13e. New York, NY: McGraw-Hill; 2011.

Right-Sided Heart Failure

Cardiac causes

Left-sided heart failure

Pulmonic valve stenosis

Right ventricular infarction

Pulmonary parenchymal diseases

Chronic obstructive pulmonary disease

Interstitial lung disease (eg, sarcoidosis)

Chronic lung infection or bronchiectasis

Pulmonary vascular diseases

Pulmonary embolism

Pulmonary arteriolar hypertension

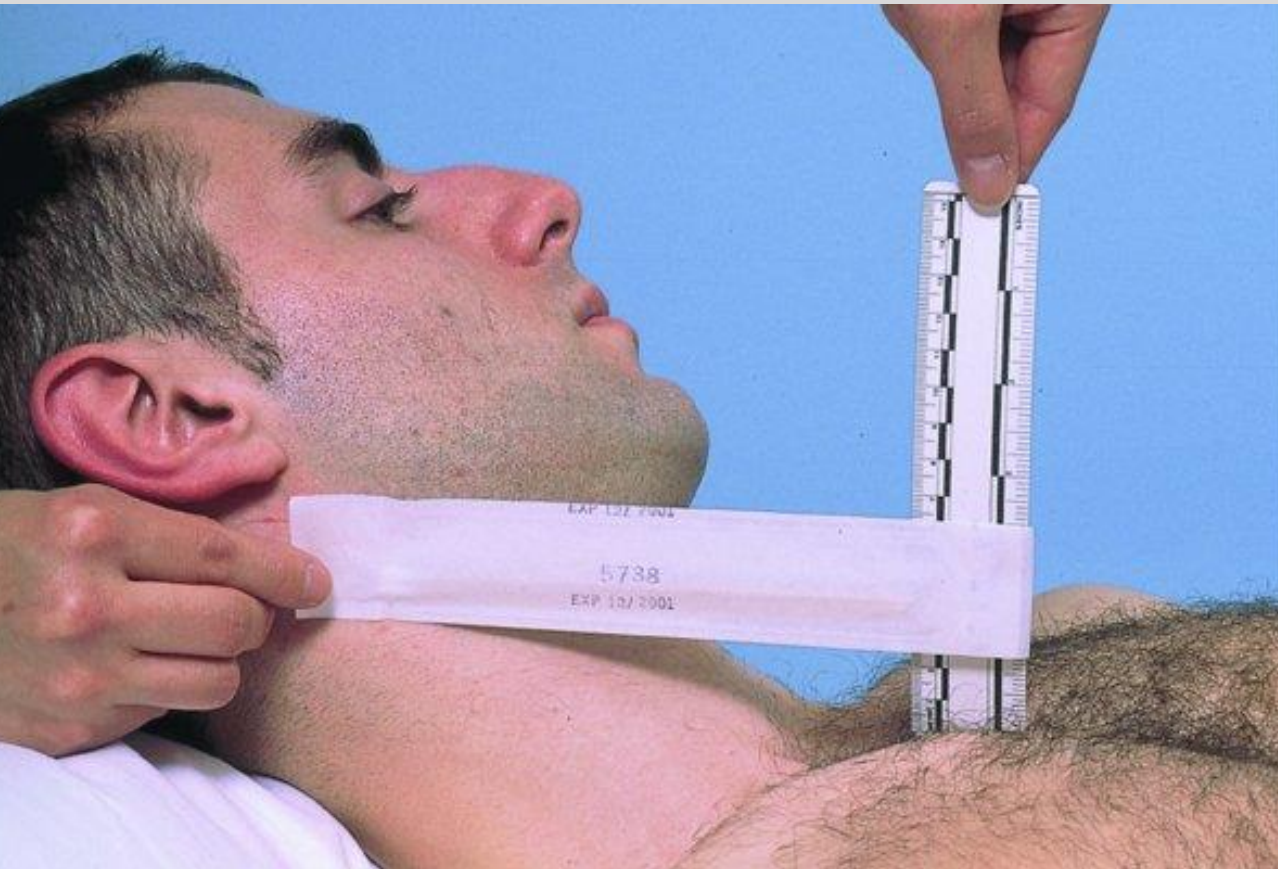
Diagnosis of Heart Failure

- **Physical Examination**
 - General Appearance and Vital Signs
 - Jugular Venous Distention
 - Pulmonary crackles (rales or crepitations)
 - Cardiac exam: displaced PMI, sustained PMI, S3 or S4, murmurs
 - Abdomen: hepatomegaly, ascites, jaundice
 - Extremities: peripheral edema
 - Sacral or scrotal edema
 - Skin: cool, indurated, pigmented extremities, cyanosis of lips and nail beds

Physical Examination

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Measure the JVP with a horizontal
card and vertical ruler



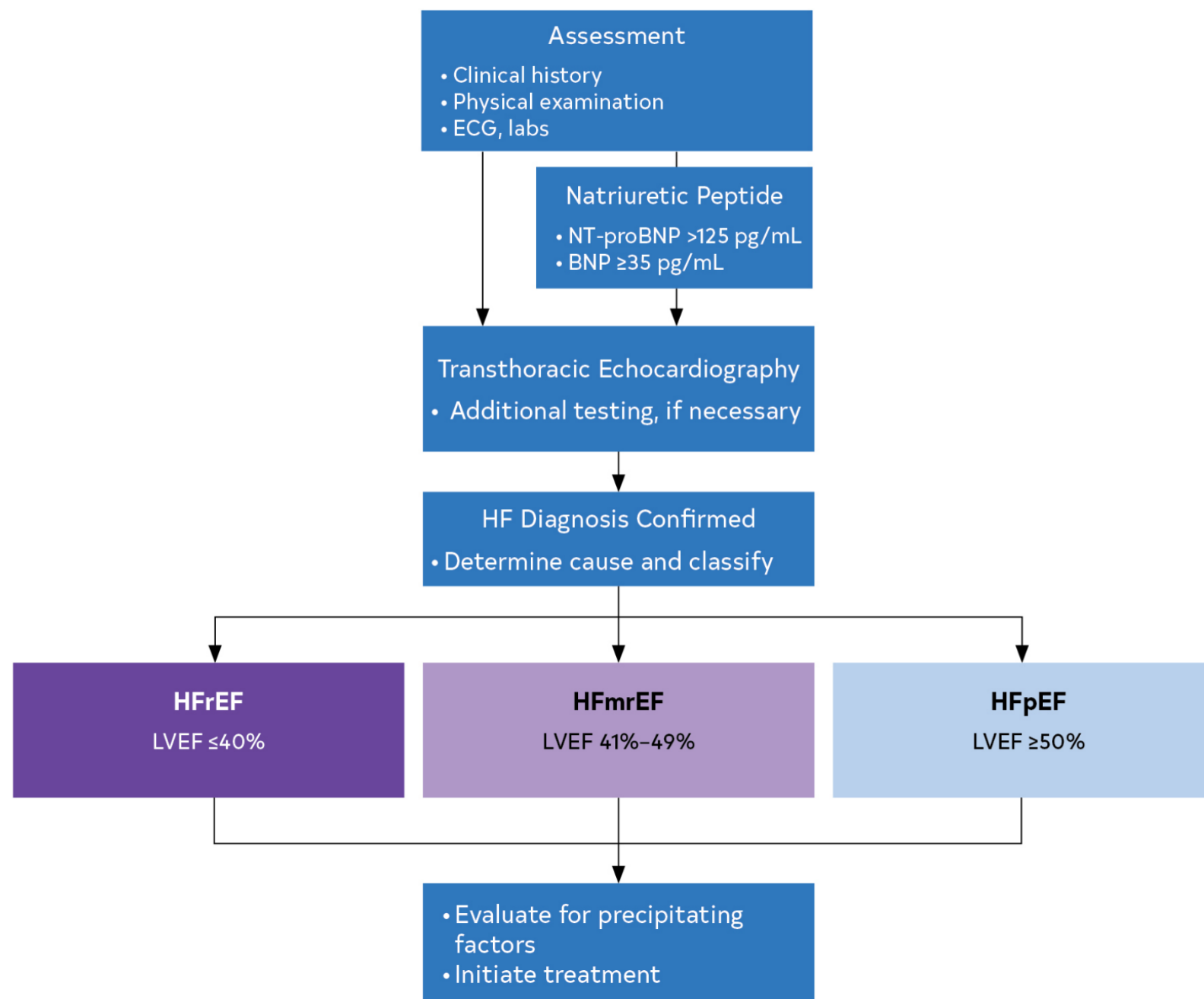
Pretibial
Edema

Diagnosis of Heart Failure

- Chest X ray
- ECG
- Labs (CBC, CMP, TFTs, HIV, lipids, A1C, COVID-19), natriuretic peptide
- Ischemic - Troponins, Coronary Angiogram
- Echo - dilated, valve disease, infiltrative cardiomyopathies
- Cardiac MRI
- Arrhythmia
- Family history
- Myocarditis
- Chronic HF with acute decompensation (all of the above plus):
 - Infection;
 - Medication or dietary non-adherence;
 - Toxins (NSAIDs, EtOH);
 - Arrhythmia

Diagnostic Algorithm for HF and EF-Based Classification

BNP indicates B-type natriuretic peptide; ECG, electrocardiogram; EF, ejection fraction; HF, heart failure; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LVEF, left ventricular ejection fraction; LV, left ventricular; NP, natriuretic peptides; and NT-proBNP, N-terminal pro-B type natriuretic peptide.



Diagnosis: Biomarkers

- **BNP (B-type natriuretic peptide)**
- **NT-proBNP (N-terminal pro-brain natriuretic peptide)**
 - Released from ventricular myocytes when ventricular diastolic pressure rises
 - Risk-stratify patients as part of the initial evaluation
 - In patients with dyspnea

- Falsely low in obesity
- Elevated in
 - Anemia
 - Advancing age
 - Renal failure
 - Women

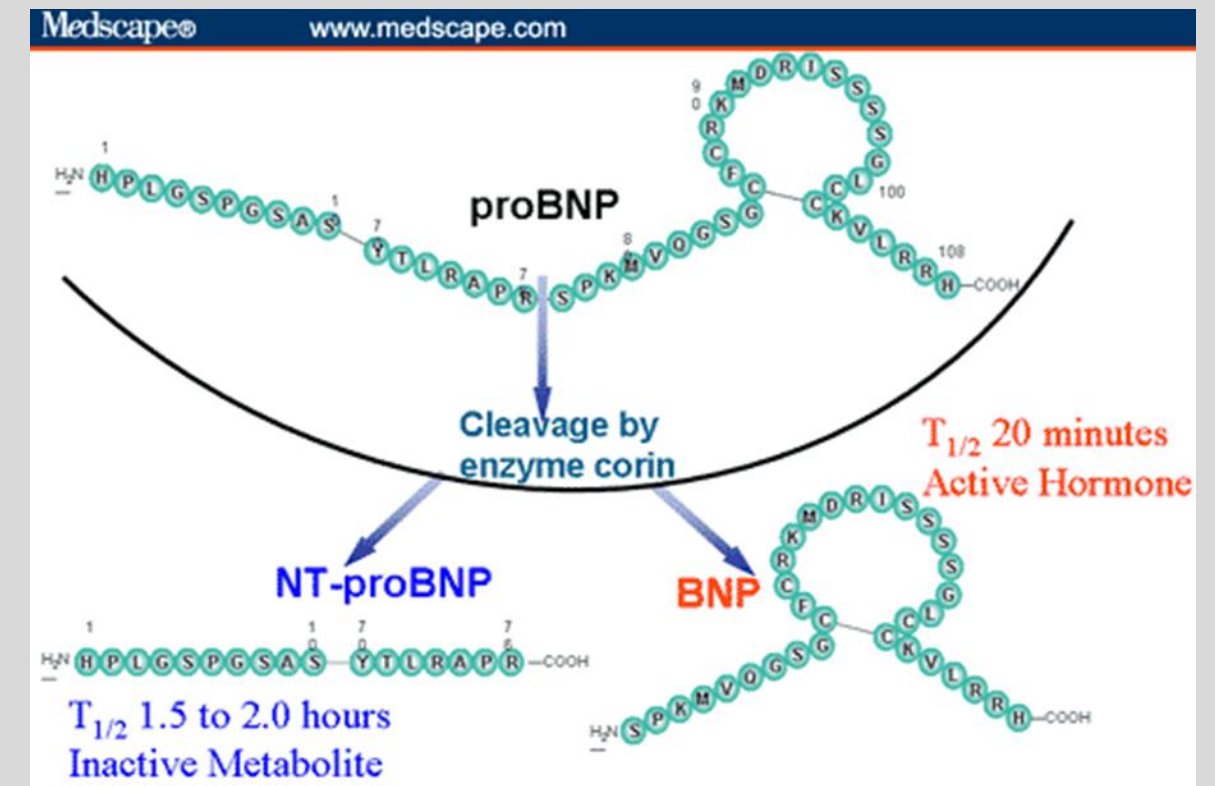


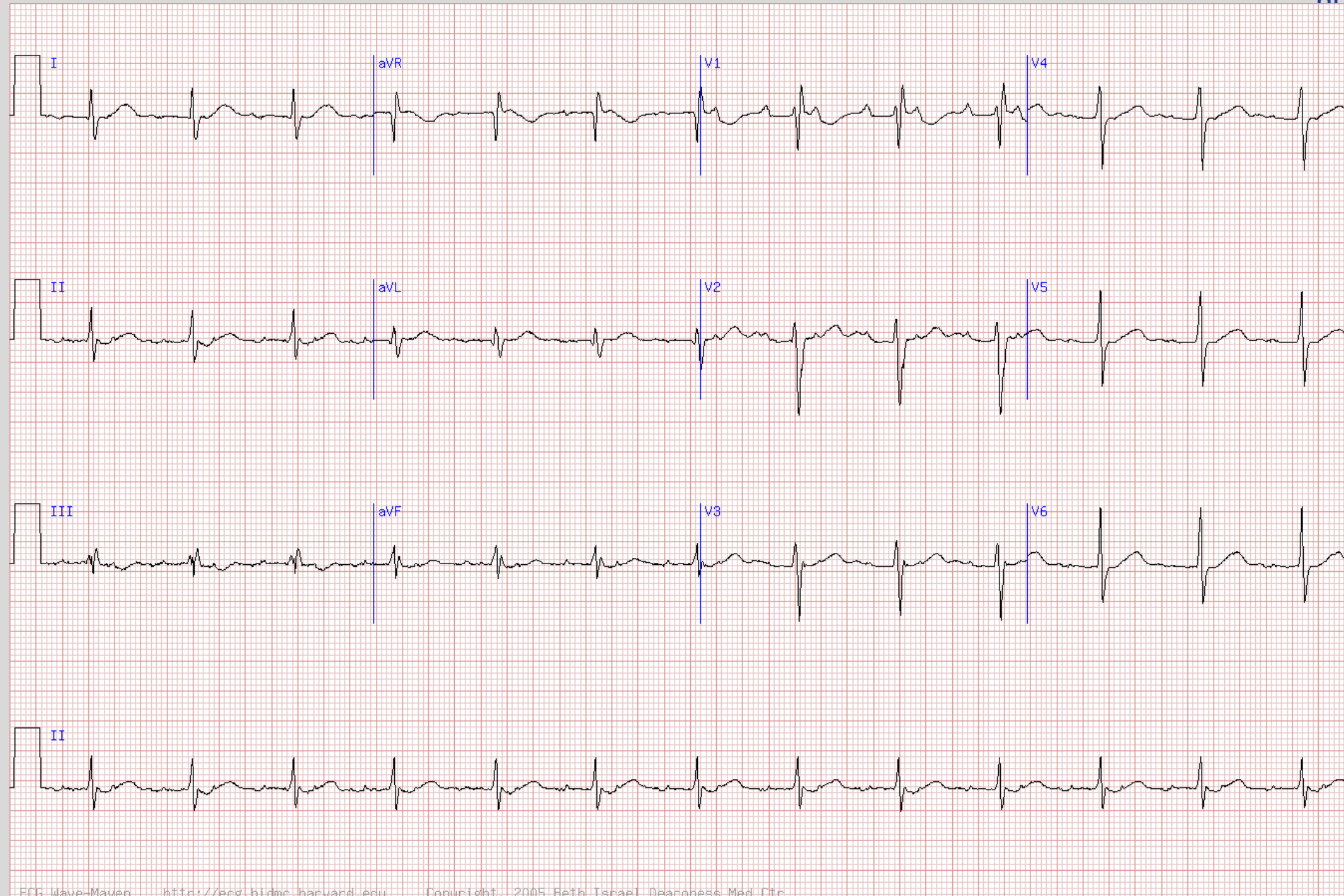
Image from <http://www.medscape.com/viewarticle/531930>

Khan M, Vaduganathan M, Butler J. Diagnosis and Management of Chronic Heart Failure. In: Fuster V, Narula J, Vaishnava P, Leon MB, Callans DJ, Rumsfeld J, Poppas A. eds. *Fuster and Hurst's The Heart*, 15e. McGraw Hill; 2022.

Diagnosis of Heart Failure: ECG #1

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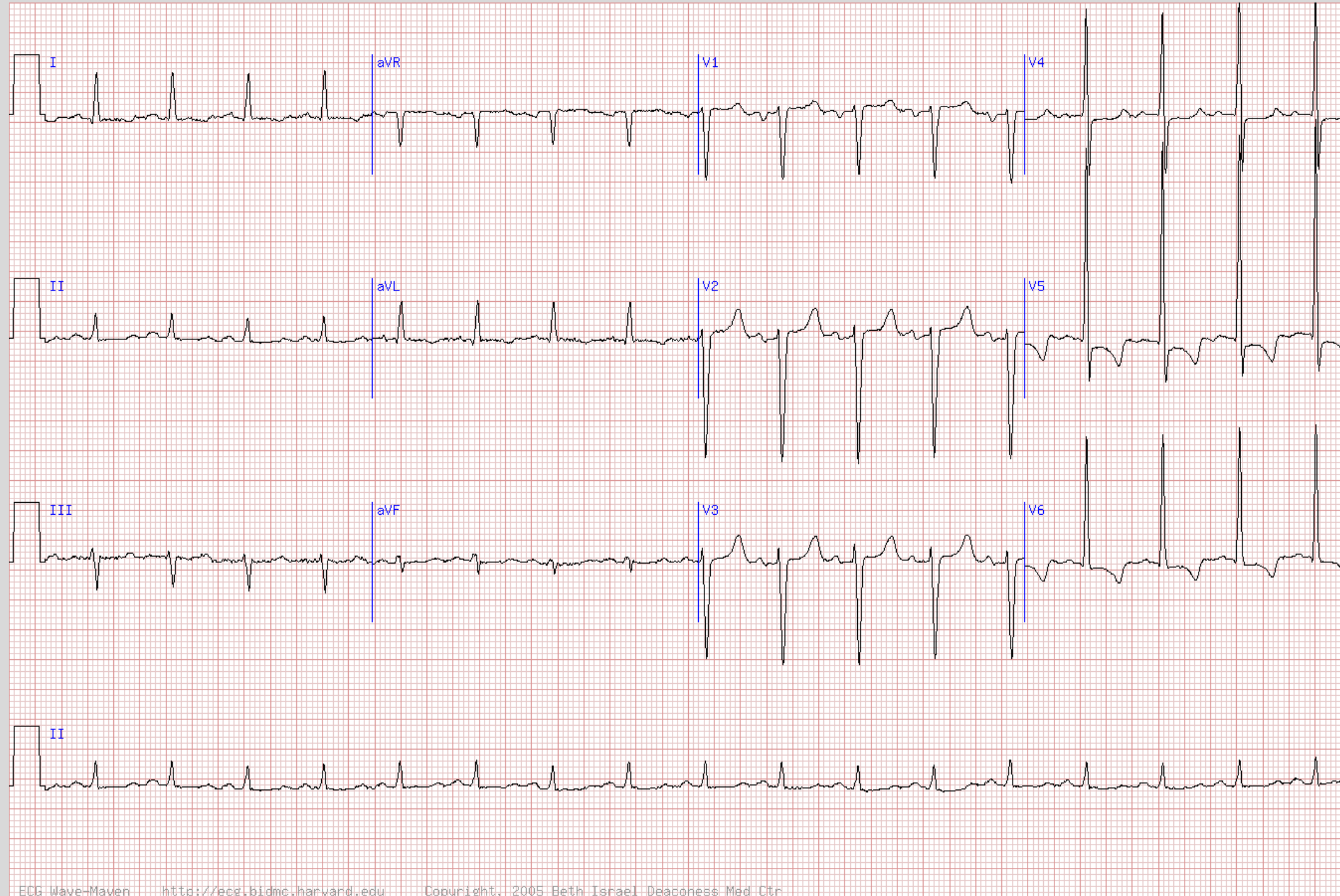
ECG Wave-Maven <http://ecg.bidmc.harvard.edu> Copyright, 2005 Beth Israel Deaconess Med Ctr

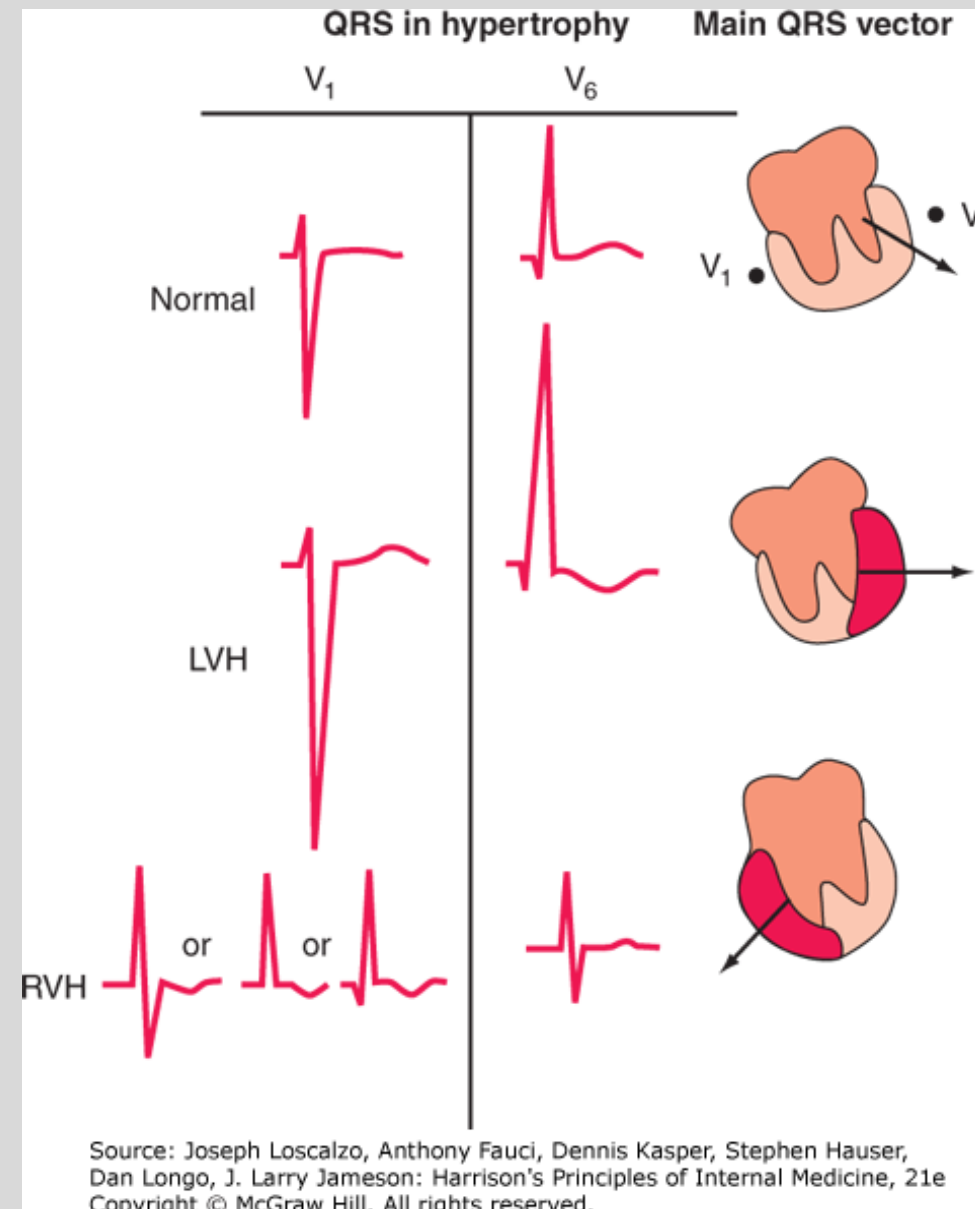
<https://ecg.bidmc.harvard.edu/maven/dispcase.asp?rownum=60&ans=1&caseid=61>

Diagnosis of Heart Failure: ECG #2

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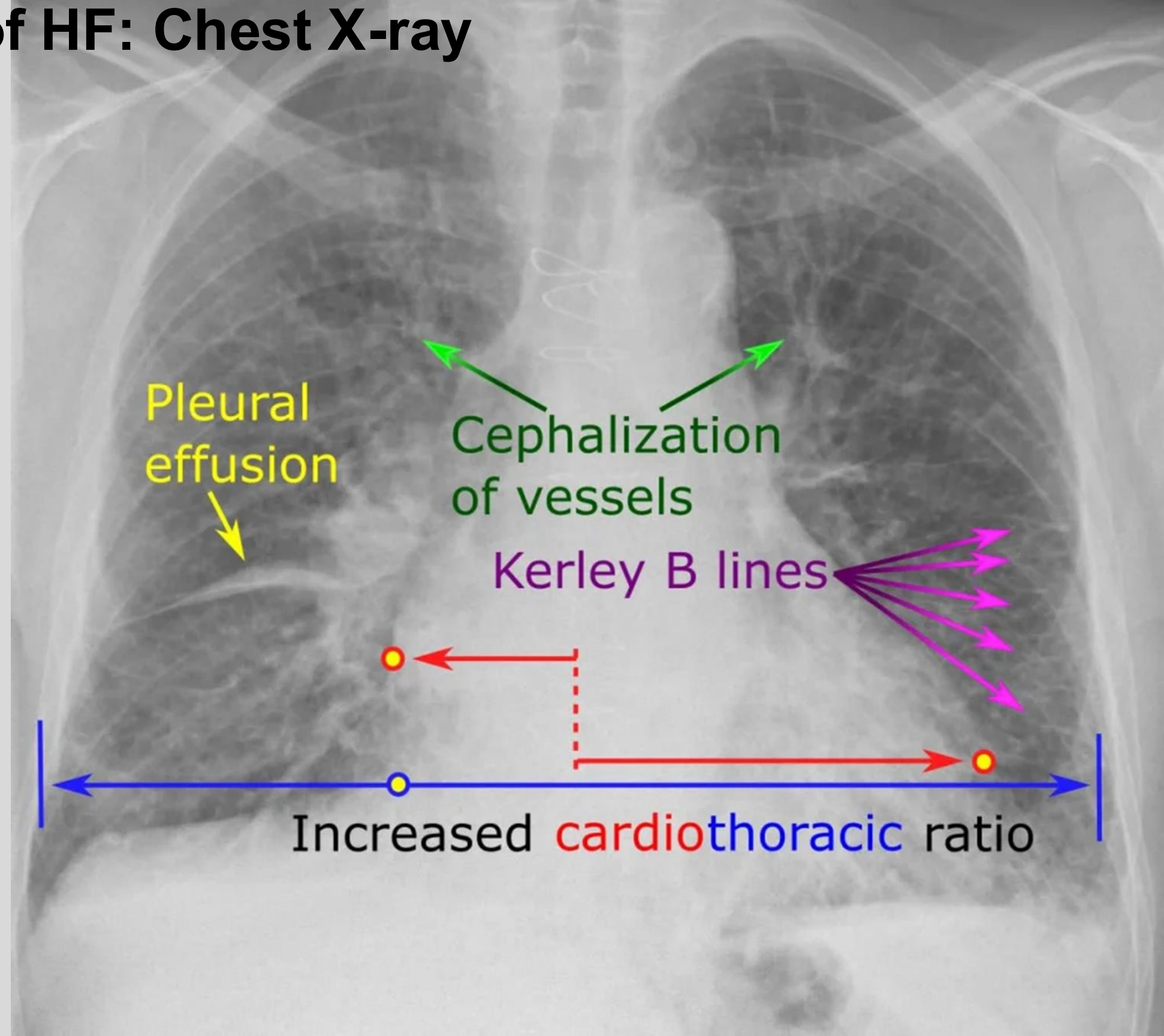




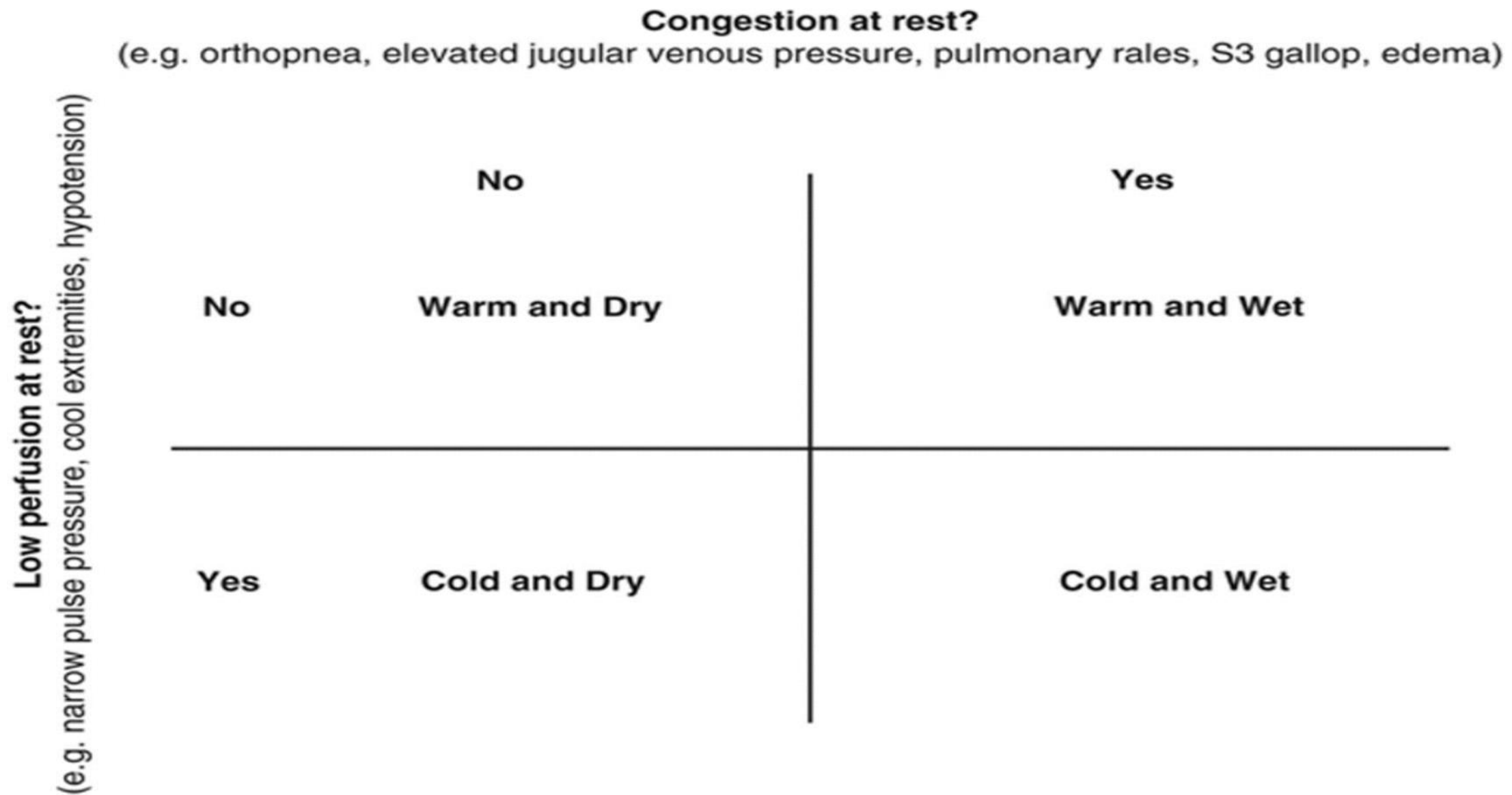
Left ventricular hypertrophy (LVH) increases the amplitude of electrical forces directed to the left and posteriorly. In addition, repolarization abnormalities may cause ST-segment depression and T-wave inversion in leads with a prominent R wave. Right ventricular hypertrophy (RVH) may shift the QRS vector to the right; this effect usually is associated with an R, RS, or qR complex in lead V₁. T-wave inversions may be present in right precordial leads.

Diagnosis of HF: Chest X-ray

- Enlarged cardiac silhouette
- Pulmonary edema
- Kerley lines are the result of interstitial edema



Acute Decompensated Heart Failure

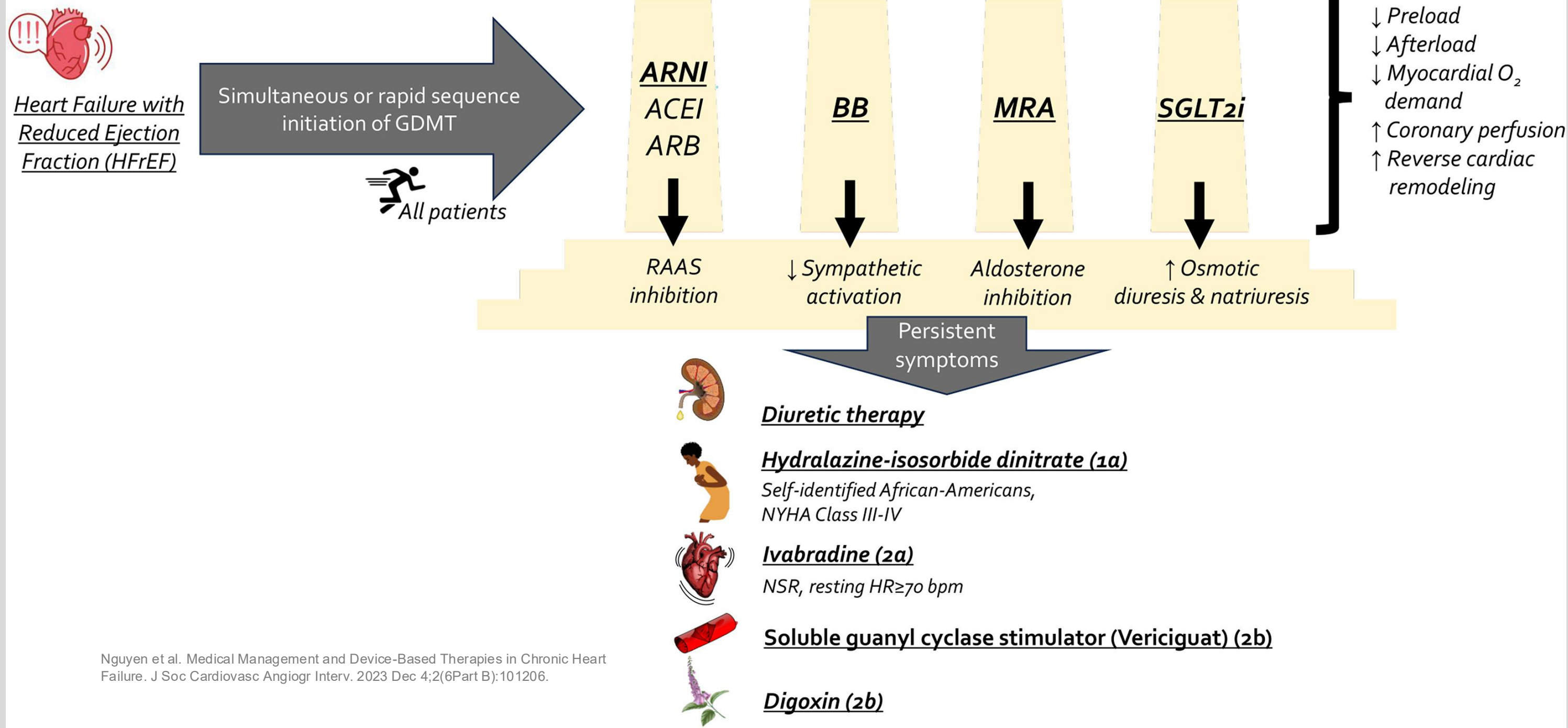


Guideline-Directed Medical Therapy

- **Pharmacologic Therapy:** HFpEF, HFpEF, HFmrEF
- **Lifestyle modifications**
 - Sodium restriction (2-3gm)
 - Fluid restriction (1.5-2L in Stage D)
 - Alcohol restriction/abstinence
 - Smoking cessation
- **Cardiac rehabilitation**
- **Device Therapy:**
 - Implantable cardioverter-defibrillator (ICD)
 - Cardiac resynchronization therapy (CRT)
- **Advanced HF Care:** Mechanical circulatory support, Transplant, Palliative care



HFrEF: Management



HFpEF: Management

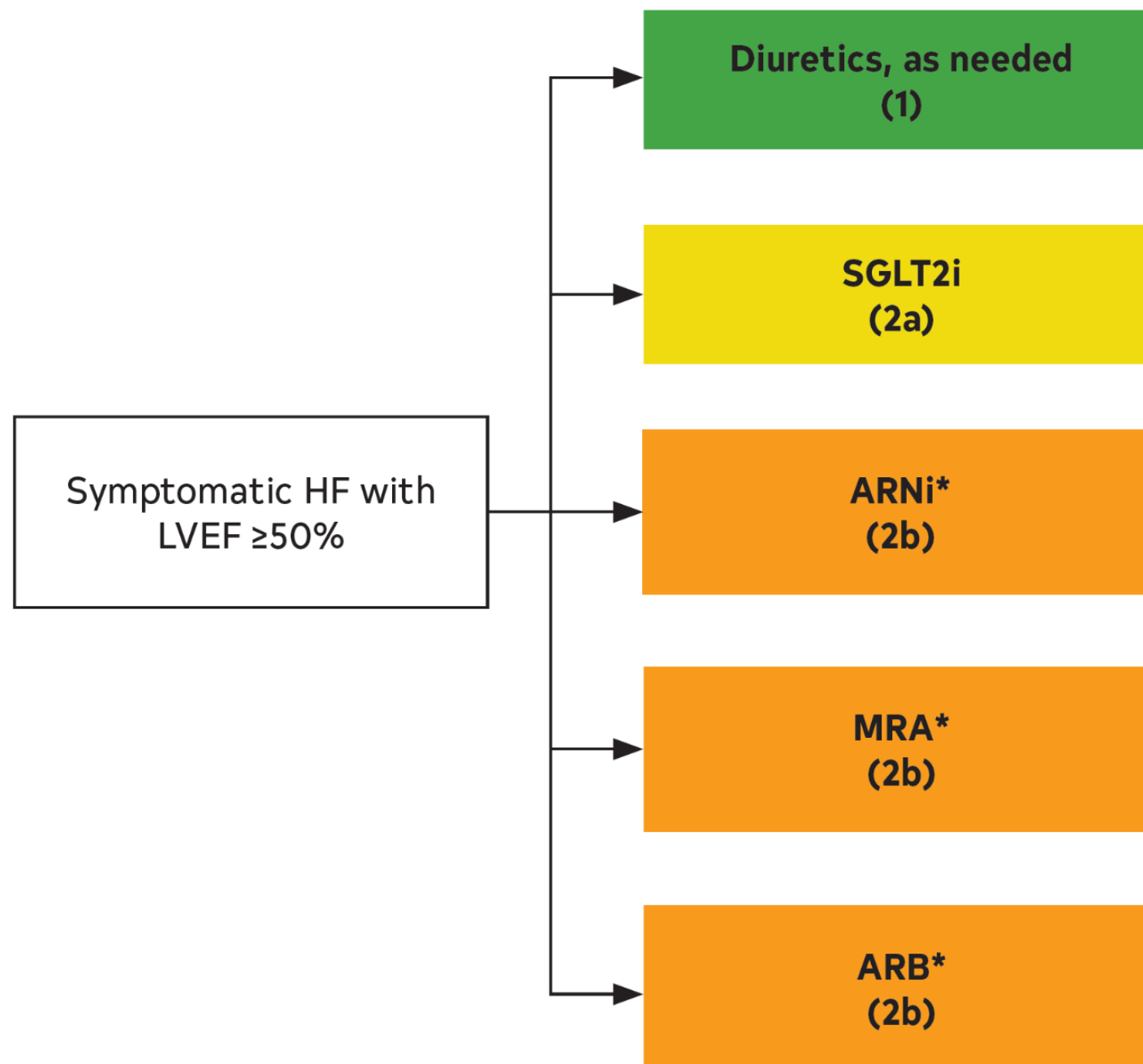
- **Treat Underlying Conditions:** HTN, diabetes, atrial fibrillation, myocardial ischemia/CAD
- **Volume Overload:** diuretics
- **Heart Rate control**
- **SGLT2 inhibitor**
- **ARNI/ACEI/ARBs**
- **GLP-1 agonist if BMI>30**
- Mineralocorticoid Receptor Antagonists (MRA)
- Beta-blockers
- Implanted Hemodynamic Monitoring
- Lifestyle modifications: diet, exercise

Recommendations for Patients With Preserved LVEF ($\geq 50\%$)

*Greater benefit in patients with LVEF closer to 50%.

ARB indicates angiotensin receptor blocker; ARNi, angiotensin receptor-neprilysin inhibitor; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; LVEF, left ventricular ejection fraction; MRA, mineralocorticoid receptor antagonist; and SGLT2i, sodium-glucose cotransporter 2 inhibitor.

Treatment of HFpEF



Acute Decompensated HF: Management

		Congestion at rest? (e.g. orthopnea, elevated jugular venous pressure, pulmonary rales, S3 gallop, edema)	
		No	Yes
Low perfusion at rest? (e.g. narrow pulse pressure, cool extremities, hypotension)	No	Optimize Oral Medications	Hypervolemia- IV Diuretics +/- Vasodilators
	Yes	Hypoperfusion: IV Fluid, Dobutamine, or Inotrope depending on PCWP	Hypoperfusion + Hypervolemia: IV Diuretics + Dobutamine, or Inotrope depending on PCWP

SODIUM-HF

Study Objective

Assess the effects of lowering the amount of sodium in the diet on clinical outcomes in a heart failure (HF) population.



Procedures



12 month intervention period
+ patients followed for additional
12 months



Dietary sodium intake assessed
using a 3-day food record.

Participants



806
Patients



26
Sites

Patients randomized to the low sodium diet followed a dietary intervention designed to reduce sodium intake to <1500 mg/day. Patients randomized to usual care followed local guidelines.

6 Countries



Results



In patients with HF, a dietary intervention to reduce dietary sodium intake did not reduce clinical events, such as being hospitalized or going to the ER.



Modest improvements were seen in quality of life and New York Heart Association HF class.



SODIUM-HF is the largest trial of its type and provides a major update to published evidence.

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sodiumhftrial.com



@Sodiumhf

SODIUM-HF

Ezekowitz JA, Colin-Ramirez E, Ross H, et al. Reduction of dietary sodium to less than 100 mmol in heart failure (Sodium-hf): an international, open-label, randomised, controlled trial. *Lancet*. 2022;399(10333):1391–1400.

GDMT: Weight Monitoring

- Daily weight
- Nurses, PT, home health staff, dietitians
- Multidisciplinary HF programs, home visiting programs, structured telephone support
- Goal “Dry” weight
- Outpatient action plan
- Sliding-scale diuretic regimen
- Intensive surveillance of weight and vital signs with use of telemonitoring has not decreased hospitalizations

Wu A. Heart Failure. Ann Intern Med. 2018;168:ITC81–ITC96. doi: 10.7326/AITC201806050
 Desai AS, Mehra MR. Heart Failure: Management. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson J. eds. *Harrison's Principles of Internal Medicine 21e*. McGraw Hill; 2022. Accessed August 07, 2022. <https://accessmedicine-mhmedical-com.arktos.nyit.edu/content.aspx?bookid=3095§ionid=265451702>



Self-Check Plan for HF Management



 **Excellent – Keep Up the Good Work!**

 No new or worsening shortness of breath
  Physical activity level is normal for you
  No new swelling, feet and legs look normal for you
  Weight check stable
Weight: ____
  No sign of chest pain

GREAT! CONTINUE:  Daily Weight Check  Meds as Directed  Low Sodium Eating  Follow-up Visits

 **Pay Attention – Use Caution!**

 Dry, hacking cough
  Worsening shortness of breath with activity
  Increased swelling of legs, feet, and ankles
  Sudden weight gain of more than 2-3 lbs in a 24 hour period (or 5 lbs in a week)
  Discomfort or swelling in the abdomen
  Trouble Sleeping

CHECK IN! Your symptoms may indicate:  A need to contact your doctor or provider  A need for a change in medications

 **Medical Alert – Warning!**

 Frequent dry, hacking cough
  Shortness of breath at rest
  Increased discomfort or swelling in the lower body
  Sudden weight gain of more than 2-3 lbs in a 24 hour period (or 5 lbs in a week)
  New or worsening dizziness, confusion, sadness or depression
  Loss of appetite
  Increased trouble sleeping; cannot lie flat

WARNING! You need to be evaluated right away.  Call your physician or call **911**



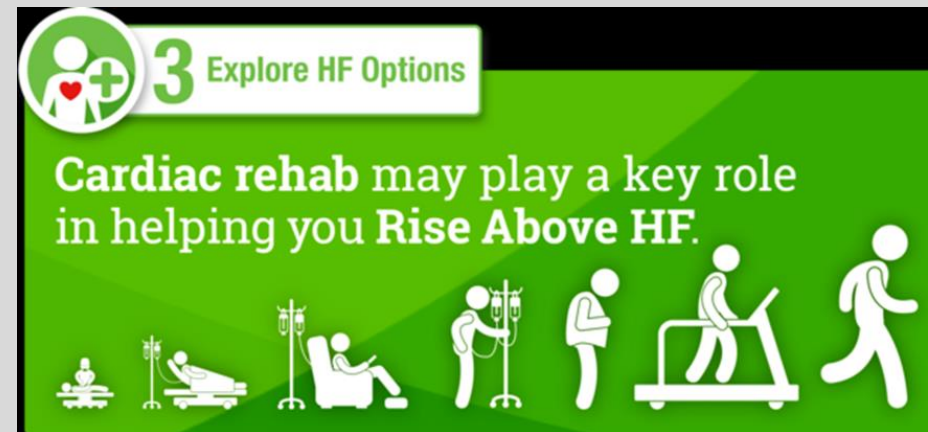
www.RiseAboveHF.org

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Nationally supported by:  **NOVARTIS** PHARMACEUTICALS

GDMT: Exercise Prescription

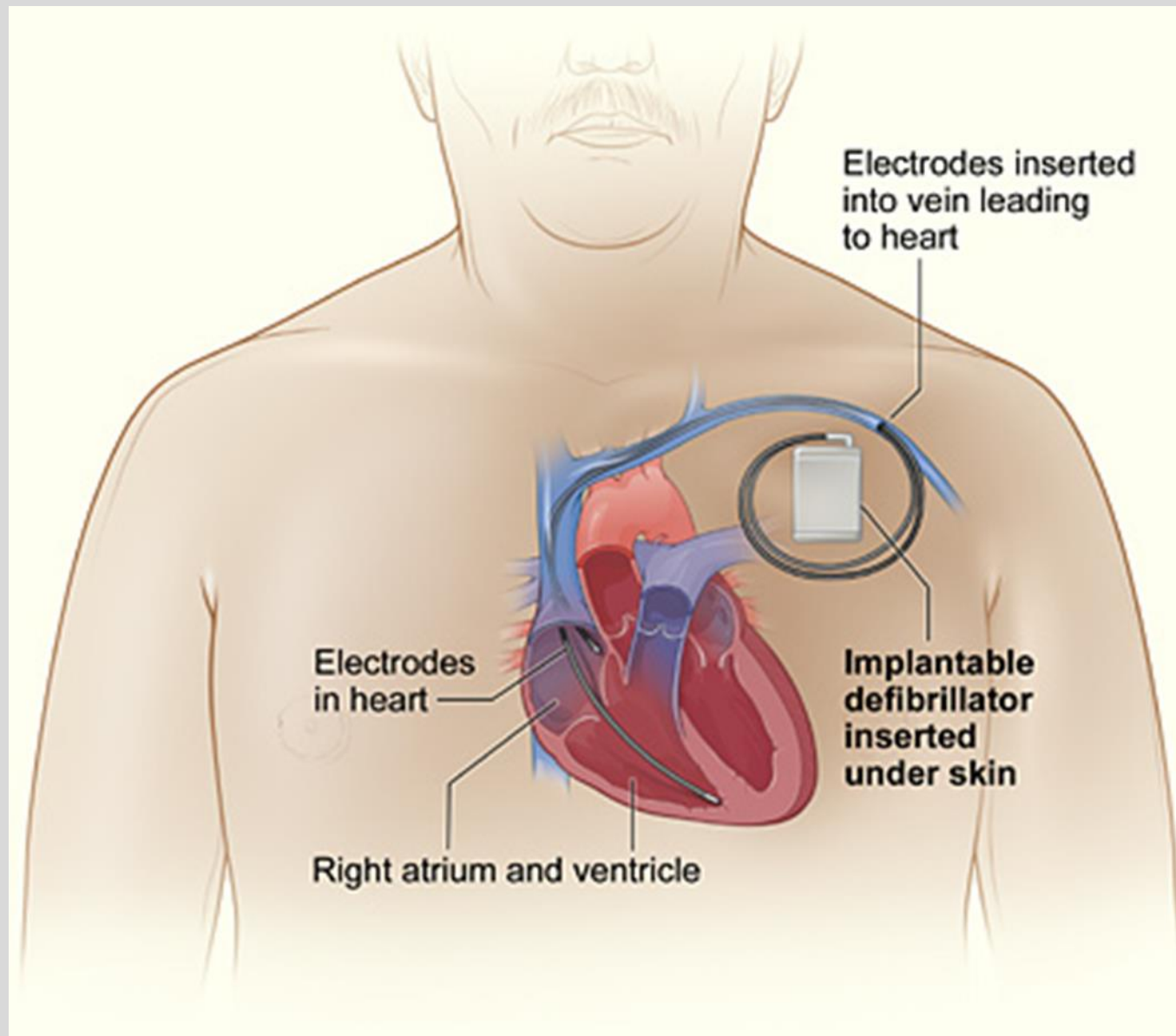
- Stable HF, NYHA II-III
- Improves functional status
- Improves exercise tolerance
- Improves quality of life
- Reduces hospitalizations
- Reduces mortality
- Contraindications: worsening HF; exercise-induced ischemia



GDMT: Implantable Cardioverter Defibrillator (ICD)

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- Prevents Sudden Cardiac Death
 - Prior sustained Vtach, Vfib;
 - Prior MI and LVEF $\leq 30\%$;
 - NYHA class II-III symptoms of HF and EF $<35\%$
- Detects and terminates life-threatening arrhythmia
- May reduce mortality in HFrEF
- ICD: Pulse generator and lead
- Biventricular ICDs (CRT-D)
- Wearable Cardioverter-Defibrillator vest

Desai AS, Mehra MR. Heart Failure: Management. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson J. eds. *Harrison's Principles of Internal Medicine*, 21e. McGraw Hill; 2022. Accessed August 11, 2022. <https://accessmedicine-mhmedical-com.arktos.nyit.edu/content.aspx?bookid=3095§ionid=265451702>

Image from <https://www.nlm.nih.gov/node/3541>

GDMT: Left Ventricular Assist Device (LVAD)

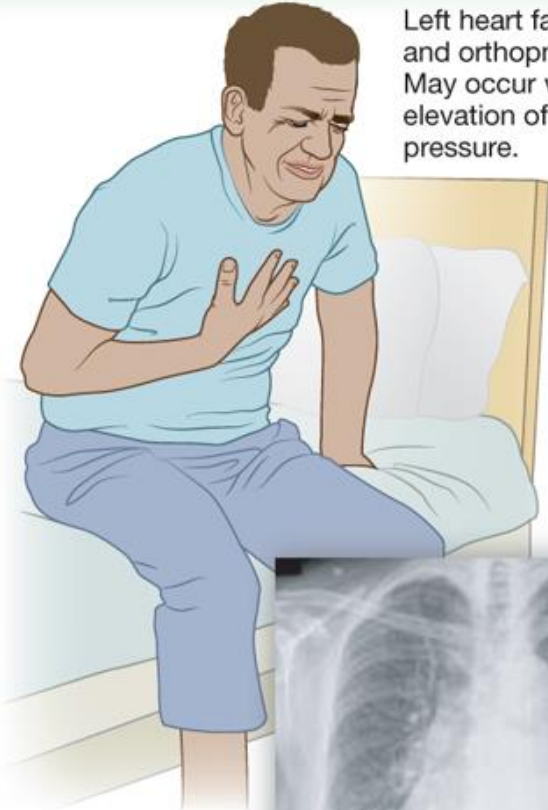
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Common transitions in advanced systolic heart failure

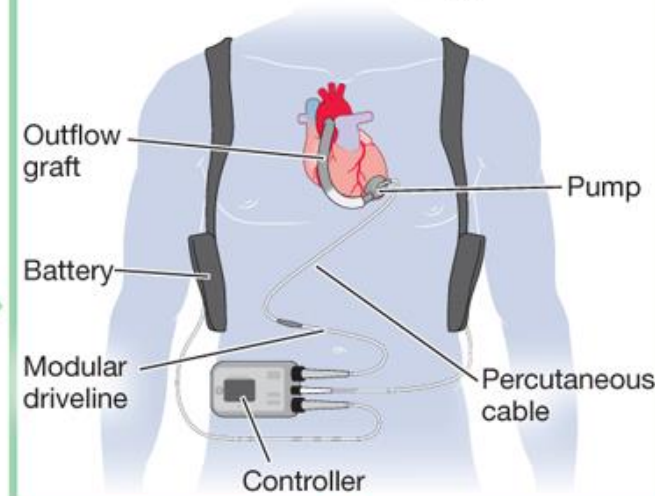
Worsening of heart failure

Left heart failure: dyspnea and orthopnea; May occur with or without elevation of central venous pressure.

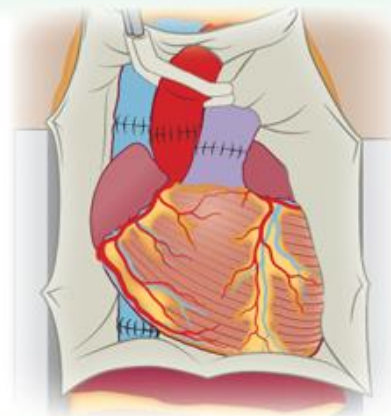


Chest x-ray: severe pulmonary congestion due to left ventricular systolic or diastolic dysfunction.

LVAD as bridge to transplant or permanent therapy



Cardiac transplantation



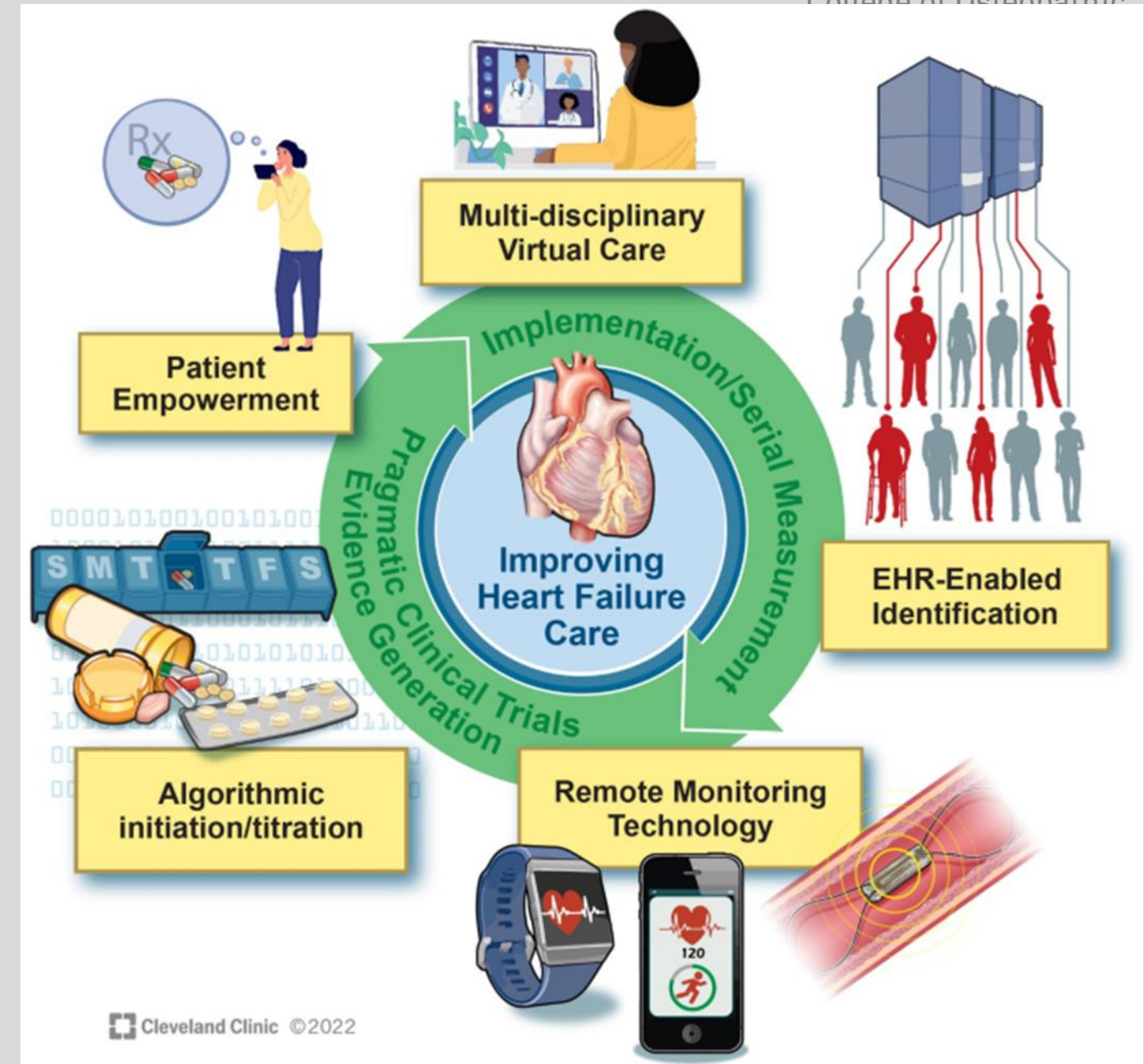
- Bridge to transplant or destination therapy
- Complications:
 - Stroke
 - Device malfunction/pump thrombosis
 - Major bleeding
 - Major infection
 - Arrhythmia

Chapter 52 Fuster and Hurst's Central Illustration. In many patients with systolic heart failure will ultimately experience worsening dyspnea, exertional intolerance, frequent hospitalizations, and end-organ dysfunction despite receiving guideline-directed medical therapy. As the disease progresses, a left ventricular assist device (LVAD) may be required to support the patient as a bridge to transplant or as permanent therapy. A subset of those with advanced heart failure will be candidates for cardiac transplantation.

Source: V. Fuster, J. Narula, P. Vaishnava, M.B. Leon, D.J. Callans, J. Rumsfeld, A. Poppas: Fuster and Hurst's The Heart, 15e
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Multidisciplinary Team Management of HF

- Referral to multidisciplinary HF disease management programs
 - Team: cardiologists, primary care clinicians, HF nurses, pharmacists, dieticians, social workers, and community health workers
- Discharge instructions
 - Activity level, diet, discharge medications, follow-up appointment, weight monitoring, cardiac rehabilitation, and what to do if symptoms worsen;
 - Details of the hospital course and the transitional plan of care (changes in medications, new diagnoses)
- Early post-discharge follow-up
 - Within 7 days of hospital discharge to optimize care and reduce rehospitalization
 - Telemedicine vs in-person, nurse/pharmacist/primary care physician



Typical List of Meds BB Clinic

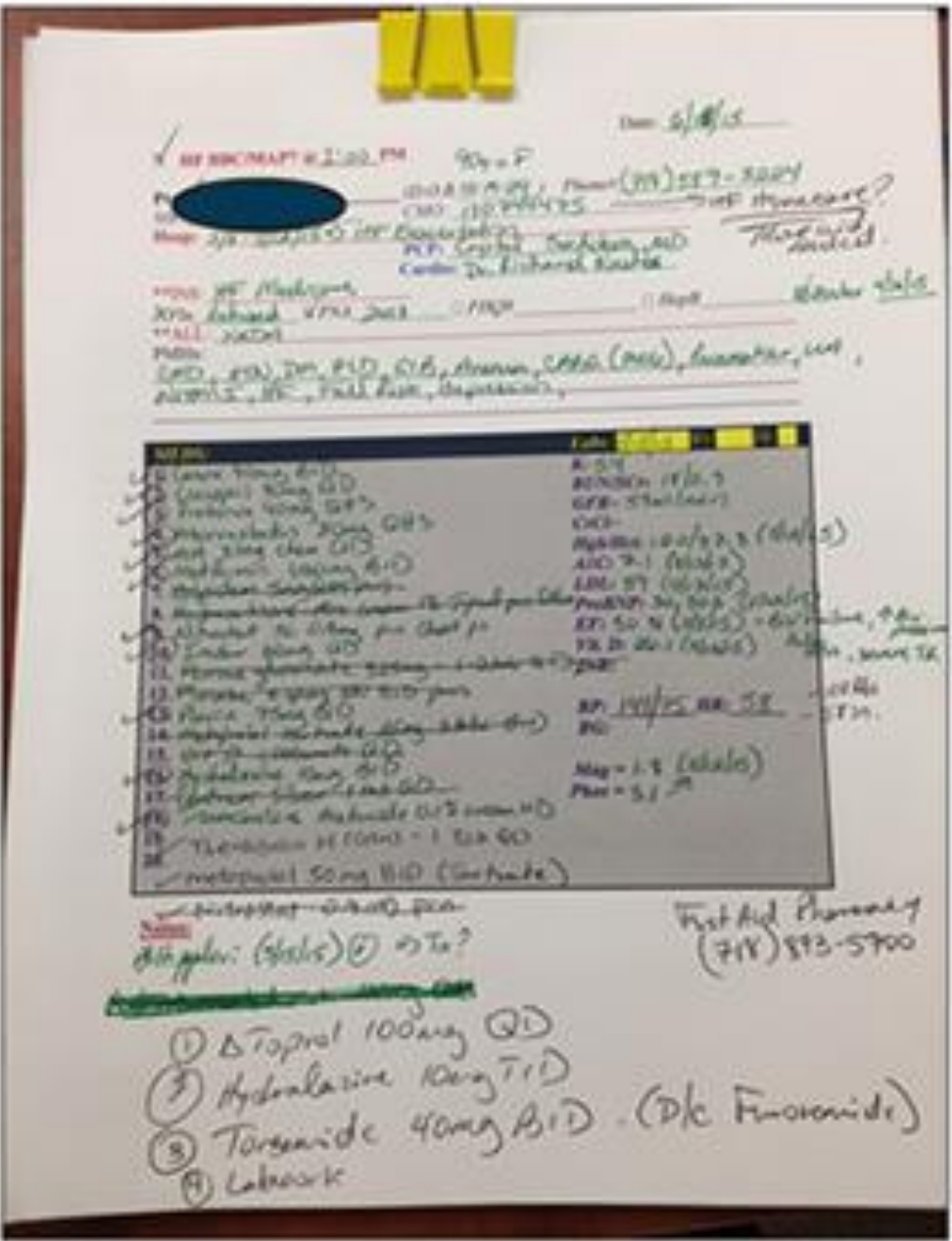
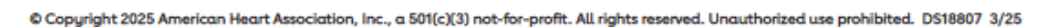


Image courtesy of Ileana Piña, MD.

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https://assets.healthfirst.org/api/pdf?id=pdf_4d70e2c035&key=50bb522a6175aced5b8a588c9d61c6e1127967b3&v=0813201606



Health Disparities in HF

- Highest incidence in self-identified Black patients (HTN)
- HF hospitalization and mortality rates for Black patients are also higher than for White
- 50,000-350,000 immigrants from Mexico and Central America may have *Trypanosoma cruzi*, with 20% progressing to Chagas cardiomyopathy
- Southeast Asian and Pacific Islanders: high prevalence of DM, associated with poor HF outcomes
- Older adults: low income, social isolation, and lack of caregiver support increase HF mortality
- Health care system factors
 - Women are less likely to receive discharge instructions for HF, less likely to be referred to specialty care, and less likely to receive a heart transplantation, compared with men;
 - Black pts are less likely to receive care from a cardiologist during an ICU admission for HF, have less access to specialized inpatient HF care, and may be vulnerable to clinician biases during evaluation for advanced HF therapies;
 - Hispanic patients are disproportionately noninsured, may experience language barriers to quality care, and also have less access to specialized inpatient HF care;
 - Native Americans and Alaskan Natives experience particular challenges in specialty care access because Indian Health Service facilities are often small and rural.

Our Patient Case (revisited)

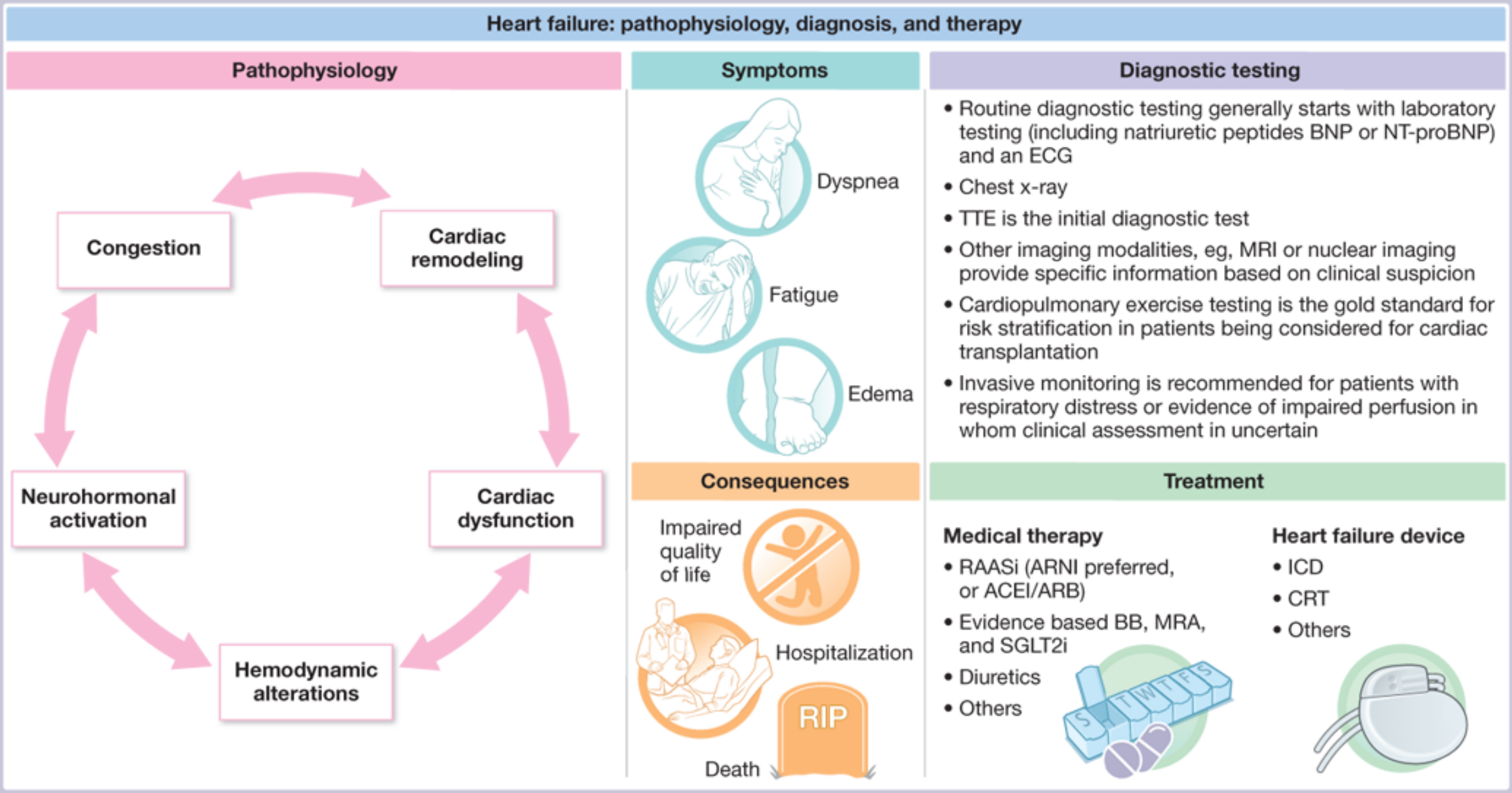
- A 65-year-old man with a history of HTN, hyperlipidemia, and tobacco use reports to the clinic because of 2 weeks of increasing dyspnea on exertion. He now sleeps on 4 pillows at night to aid in breathing. He has T2DM variably controlled with metformin and diet, and drinks 3-4 glasses of wine/day. Physical examination: jugular venous distention, an S_3 heart sound, and bilateral LE edema. ECG shows Q waves in leads V_2 - V_4 . He gives a history of hospitalization for "issues with my heart" about a year ago when he had sudden chest pain that radiated to his left arm, after which he had been asymptomatic.
- Which of the following is the most likely cause of this patient's condition?
 - A. Alcohol
 - B. Aortic stenosis
 - C. Coronary artery disease
 - D. Coronavirus
 - E. Coxsackievirus B
 - F. Sarcomere gene mutation



Summary

- HF is a complex clinical syndrome with symptoms and signs that result from any structural or functional impairment of ventricular filling or ejection of blood.
- Two major types of HF are heart failure with reduced ejection fraction or heart failure with preserved ejection fraction.
- Classic symptoms of acute HF: exertional dyspnea, paroxysmal nocturnal dyspnea, orthopnea, and peripheral edema.
- Guideline-directed medical therapy for HFrEF: 4 medication classes (ARNI, beta-blockers, MRA, SGLT2i) and diuretics for volume overload.
- Core References:
 - Givertz MM, Mehra MR. Givertz M.M., & Mehra M.R. Givertz, Michael M., and Mandeep R. Mehra. Heart Failure: Pathophysiology and Diagnosis. In: Longo DL, Fauci AS, Kasper DL, Hauser SL, Jameson J, Loscalzo J, Holland SM, Langford CA. Longo D.L., & Fauci A.S., & Kasper D.L., & Hauser S.L., & Jameson J, & Loscalzo J, & Holland S.M., & Langford C.A.(Eds.),Eds. Dan L. Longo, et al.eds. *Harrison's Principles of Internal Medicine, 22nd Edition*. McGraw Hill; 2025. Accessed July 01, 2026. <https://nyit.idm.oclc.org/login?url=https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3541§ionid=296165244>

Summary



Chapter 48 Fuster and Hurst's Central Illustration. Heart failure is a clinical syndrome that involves a complex interplay between myocardial, vascular, neurohormonal, hemodynamic and comorbid factors. Signs and symptoms of heart failure can be fairly nonspecific and thus diagnosis of heart failure is often challenging. However, early diagnosis and treatment of heart failure can improve quality of life, and reduce risk of hospitalization and/or death. BB, beta blockers; BNP, B-type natriuretic peptide; CRT, cardiac resynchronization therapy; ECG, electrocardiogram; ICD, implantable cardioverter defibrillator; MRA, mineralocorticoid receptor antagonist; NT-proBNP, N-terminal-pro B-type natriuretic peptide; RAASi, renin-angiotensin-aldosterone system inhibitors; SGLT2i, sodium-glucose co-transporter 2 inhibitors; TTE, transthoracic echocardiography.

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Thank you!